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A Mobile LiDAR-Based Deep Learning Approach for Real-Time 3D Body Measurement

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Abstract: In this study, we propose a solution for automatically measuring body circumferences by utilizing the built-in LiDAR sensor in mobile devices. Traditional body measurement methods mainly rely on 2D images or manual measurements. This research, however, utilizes 3D depth information to enhance both accuracy and efficiency. By employing HRNet-based keypoint detection and transfer learning through deep learning, the precise locations of body parts are identified and combined with depth maps to automatically calculate body circumferences. Experimental results demonstrate that the proposed method exhibits a relative error of up to 8% for major body parts such as waist, chest, hip, and buttock circumferences, with waist and buttock measurements recording low error rates below 4%. Although some models showed error rates of 7.8% and 7.4% in hip circumference measurements, this was attributed to the complexity of 3D structures and the challenges in selecting keypoint locations. Additionally, the use of depth map-based keypoint correction and regression analysis significantly improved accuracy compared to conventional 2D-based measurement methods. The real-time processing speed was also excellent, ensuring stable performance across various body types.

Keywords: LiDAR; HRNet; deep learning; keypoint



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1. Introduction

The fourth industrial revolution is bringing about rapid changes across various industries through the convergence of innovative technologies such as Artificial Intelligence (AI), Big Data, the Internet of Things (IoT), and robotics. These technological advancements assist human tasks, create new business models, and contribute to maximizing overall societal efficiency. In particular, the field of computer vision, integrated with deep learning-based AI technologies, is opening up diverse application possibilities by enabling computers to understand and interpret visual information through images or videos. Computer vision is a technology that allows computers to analyze and recognize visual data similarly to humans by utilizing advanced techniques such as machine learning, deep learning, and image processing. This field is divided into various subfields, including object recognition, facial recognition, image segmentation [1], and motion recognition, and is employed as a core technology in numerous industries such as autonomous driving, security systems, medical image analysis [2], and sports analytics [3–5].

For example, object recognition technology [6] is essential for autonomous vehicles to accurately recognize vehicles, pedestrians, and traffic lights on the road, while facial recognition technology is used in security systems for identity verification and access

control. Additionally, image segmentation technology [7] is utilized to accurately identify and analyze tumors or other lesions in medical imaging, and motion recognition technology is applied in analyzing the movements of athletes or in human behavior recognition systems. However, these computer vision technologies are primarily based on 2D images or video data as shown in the Figure 1, and accurate length measurement solutions in 3D space have not yet been sufficiently developed. 3D length measurement can play a significant role in various fields such as architecture, manufacturing, healthcare, and fashion, and can greatly enhance accuracy and efficiency in personalized services like body size measurement.

In recent years, LiDAR (Light Detection And Ranging) [8] technology has rapidly advanced by expanding its range of applications. LiDAR is a technology that calculates distances by emitting lasers and measuring the time it takes for the reflected light from objects to return, thereby generating high-resolution 3D point clouds. This technology is utilized in various fields such as autonomous vehicles, drones, robotics, and Geographic Information Systems (GIS), and it particularly excels in precise data collection and analysis in 3D spaces. Additionally, LiDAR sensors embedded in mobile devices provide high-resolution depth information in consumer-grade devices, enabling general users to easily utilize 3D data. LiDAR sensors can operate in both vertical and horizontal directions and can create precise 3D models of objects through multi-directional scanning. Due to these advantages, LiDAR is extensively applied in specialized fields such as surveying, geomorphology, archaeology, laser guidance, and high-resolution mapping. In particular, high-resolution 3D scanning has established itself as an essential tool for precisely analyzing complex terrains and structures. However, up to now, LiDAR technology has been primarily used in specialized equipment or industrial settings, and its utilization in everyday life has been limited. This is due to the high cost of equipment and complex maintenance requirements, making it perceived as a technology that is difficult for general users to access. Therefore, developing a more affordable and user-friendly 3D measurement solution by leveraging LiDAR sensors embedded in mobile devices has become an important research task. One of the application areas for this technology, body size measurement, plays a crucial role in various fields such as the apparel industry, healthcare, and fitness. Traditionally, body measurements have been carried out in a manual manner, which can be time-consuming and inconvenient for users. Especially when purchasing clothing online, accurately measuring body sizes is essential, but until now, there has been the inconvenience of users having to measure their dimensions using measuring tools themselves. To address these issues, the development of automatic body size measurement technology is necessary.

The goal is to develop 3D length measurement technology by utilizing the standard 2D RGB camera and built-in LiDAR sensor of mobile devices to extract depth information alongside 2D images of subjects. Through this, we seek to provide a platform that allows easy measurement of body dimensions in everyday life without the need for specialized 3D equipment. Software utilizing mobile devices can be implemented at a low cost and can significantly enhance user convenience by measuring lengths in real time and providing immediate results. Additionally, an intuitive user interface increases accessibility to the technology and efficiently uses energy and resources, thereby reducing time and energy consumption. Specifically, body size is an important metric in clothing purchases. Previously, users had to measure their dimensions themselves, but the automatic dimension measurement platform proposed in this study supports users in easily measuring their body dimensions, thereby greatly enhancing the convenience of online shopping. This is expected to improve user experience and have positive effects such as reducing return rates.

This study makes several key contributions. First, it developed a technology for accurately measuring 3D lengths by utilizing the LiDAR sensor embedded in mobile devices without relying on specialized 3D equipment. Second, it introduced a keypoint

detection technique based on the HRNet model and improved measurement accuracy by accurately identifying the positions of body parts through transfer learning. Third, it developed an algorithm that effectively integrates 2D RGB images and 3D depth maps, enabling accurate length measurements in three-dimensional space. Lastly, it implemented an algorithm that automatically measures the circumferences of body parts based on images captured from the front and side, and demonstrated high accuracy through experiments.



Figure 1. An example of human pose estimation using 2D images.

Meanwhile, there has also been extensive research on estimating body circumference by reconstructing the human body as a 3D mesh. Approaches such as SMPL [9], PIFu [10], and DeepHuman [11] share a similar concept with this study in that they attempt to restore a three-dimensional shape from limited images (or scanned data). However, each model differs in problem setting and application context, requiring specialized equipment, complex computational procedures, or parametric learning based on a naked body model. In this study, we discuss these methods and highlight their differences from our approach more specifically in Section 2.4.

This paper is structured as follows. Section 2 reviews related studies and discusses the distinctions from the current research. Section 3 provides a detailed explanation of the principles behind storing 3D depth information using a mobile phone LiDAR sensor. Section 4 introduces the proposed methodology, detailing the processes of data collection, preprocessing, keypoint detection and merging, and the body circumference measurement algorithm. Section 5 presents the experimental setup and results, validating the effectiveness of the proposed method through result analysis. Finally, Section 6 discusses the limitations of this study and outlines directions for future research.

2. Mobile Device 3D Measurement Technology

As shown in Figure 1, research on body measurement has primarily been based on 2D image analysis. These studies have focused on methods that detect the positions of body parts using pose estimation techniques and calculate circumferences based on them. In particular, high-resolution pose estimation models like HRNet are effective in accurate keypoint detection, but due to the limitations of 2D images, 3D depth information is not reflected, restricting the accuracy of measurements [12]. Research on 2D image-based body measurement has primarily relied on pose estimation and keypoint detection. The HRNet model excels at accurately detecting keypoints of body parts while maintaining high-resolution representations, allowing precise identification of the locations of various body parts. However, these methods have limitations in that they do not consider depth information, thereby failing to accurately reflect the 3D structure of the body. For example,

they may not account for depth variations of the body or distortions due to various postures, which can reduce measurement accuracy [13]. Additionally, [14] proposed a method to track body movements through real-time pose estimation, but it is primarily focused on motion analysis and is not suitable for static body circumference measurements [15]. Attempted 3D pose estimation, but still did not fully incorporate depth information from 2D images, resulting in limitations in accuracy.

2.1. 3D Measurement Technologies and Applications of LiDAR

3D measurement technologies overcome the limitations of 2D images and enable more accurate body measurements. Representative 3D measurement technologies include Structured Light [16], Stereo Vision [17], and LiDAR [8]. Among these, LiDAR is a technology that can generate high-resolution 3D point clouds by measuring distances to objects using lasers, and is widely utilized in various fields such as autonomous vehicles, robotics, and Geographic Information Systems (GIS) [18]. LiDAR technology has several advantages. It allows for highly accurate distance measurements through laser emission and reflection, can generate a large number of point clouds at high speeds enabling real-time data processing, and can be utilized in various fields such as construction, surveying, archaeology, and environmental monitoring [19]. However, in the field of body measurement, the application of LiDAR is relatively insufficient. This is mainly due to the need for specialized equipment, high costs, and complex data processing requirements, making it perceived as a technology that is difficult for general users to access [20,21]. Although some recent studies have proposed body recognition and measurement methods using LiDAR data, these studies remain mostly at the research stage, and implementation as practical solutions is limited [22].

2.2. LiDAR-Based Body Measurement Technology

Recently, research on body measurement utilizing LiDAR sensors has been increasing. For example, ref. [22] proposed a method to automatically recognize and measure human body parts using LiDAR data. In this study, 3D point clouds from LiDAR were used to accurately identify various body parts and measure body circumferences based on them. However, this method primarily assumes measurements in a fixed environment, limiting its applicability in various everyday situations. Another study [23] proposed a method to predict the locations of body parts based on LiDAR data using a deep learning model. This study effectively utilized depth information from LiDAR to create a 3D body model, showing higher accuracy than conventional 2D image-based methods, but research on real-time processing and user-friendly interface implementation is still lacking.

2.3. Data Collection for 3D Measurement

In this study, uniform 3D data, such as cylindrical object and mannequin, were used to create initial datasets and later expanded to actual human body measurements. The experiments began with relatively simple models that resemble specific human body parts to minimize deformation, enhance the accuracy of the measurement algorithm, and enable a systematic evaluation across various body parts. Notably, previous studies such as [22,23] primarily conducted experiments on real human subjects. In contrast, this study verified the performance of the algorithm in the early stages using uniform 3D objects. This approach helped secure fundamental accuracy and minimize variables that could arise when measuring different body parts. Additionally, the use of cylindrical object and mannequin ensured data consistency, which played a critical role in the training and evaluation processes of the deep learning model.

In this study, we first verified the basic performance of the measurement algorithm using uniform 3D data, such as cylindrical objects and mannequins, and then extended

our experiments to actual human subjects. A total of 150 Korean male participants were involved, each wearing black tights during the experiment to minimize potential errors caused by garment wrinkles or reflections.

- Four Captures (Front/Side, Upper/Lower Body): We divided the upper and lower body and conducted a total of four captures—one each from the front and side views.
- Front View: Participants stood with their arms extended to the sides and feet slightly apart, ensuring the torso outline was clearly visible, while capturing RGB images and LiDAR depth maps simultaneously.
- Side View: Participants extended their arms forward to facilitate accurate chest circumference measurement. Again, RGB and LiDAR depth maps were collected in the same manner. Lighting, camera distance, and angles were kept consistent, and if a large discrepancy was found by comparing the results with ground-truth measurements over two or more captures, a retake was performed.

We recruited participants with waist circumferences from below 72 cm (approximately 11% of the subjects, considered slim) to above 92 cm (about 16% of the subjects), thereby confirming that the proposed algorithm can function reliably over a broad range of body sizes. Each participant underwent four total captures (front/side, upper/lower body), yielding both RGB and depth map sets for a detailed evaluation of full-body circumference measurement.

2.4. Limitations of Existing Studies and Differentiation of This Study

Recent studies have reported attempts to leverage 3D data acquired from LiDAR sensors for body circumference measurement. However, despite the active investigation of 3D mesh reconstruction-based methods for human body measurements, models such as SMPL, PIFu, and DeepHuman still face structural limitations for immediate circumference measurement in mobile environments. SMPL, for instance, is a parametric model trained on nude body shapes, making it difficult to fully account for variations in body shape and circumference when clothing is involved. Additionally, estimating SMPL parameters (e.g., β , θ) from a single image demands a complex fitting procedure requiring a high-performance GPU, rendering real-time application practically infeasible.

Meanwhile, PIFu can reconstruct high-resolution 3D surfaces using implicit functions such as capable of capturing fine details such as clothing wrinkles, but accurately estimating the rear or occluded regions from a single viewpoint remains difficult, and the method typically requires multi-view acquisition and substantial computational resources, making it unsuitable for typical mobile devices. Finally, DeepHuman employs a voxel-based 3D reconstruction approach, where increasing resolution leads to exponential growth in GPU memory usage and computational overhead. Moreover, because the model must infer unobservable areas from a single viewpoint, the margin of error increases. Ultimately, without multi-view input, achieving sufficient accuracy is challenging, and near-real-time operation on mobile devices also remains limited.

To overcome the computational and practical constraints of these 3D mesh reconstruction approaches, this study proposes a lightweight method that combines 2D keypoint detection with LiDAR-sensor data. Specifically, we rapidly identify (target) the measurement regions using a high-resolution 2D pose estimation network, such as HRNet, and then compute only the simple Euclidean distance in 3D space. This method eliminates the need for complex 3D mesh reconstruction while still leveraging depth data, allowing for accurate circumference measurements with lower computational cost. As a result, it provides sufficiently accurate measurements from a single viewpoint and can operate in near-real time on mobile devices without the need for high-performance GPU or multi-view data.

Experimentally, we validated our algorithm's performance using uniform 3D data such as cylindrical objects and mannequins, demonstrating high accuracy and practical utility across various body regions. In particular, on a system equipped with an AMD Ryzen 5950X (16 cores/32 threads), 128 GB of RAM, two RTX 3090 24 GB GPUs, and a 1 TB M.2 SSD, keypoint detection and circumference measurement took an average of 0.7 s, indicating near-real-time processing capability. This underscores the advantage of our lightweight approach, which relies solely on CPU-based Euclidean distance computation following 2D keypoint detection, eliminating the need to reconstruct the entire 3D space.

3. Proposed Algorithm for Mobile 3D Body Circumference Measurement

3.1. Merging Keypoints and 3D Data

In this study, the overall process of body circumference measurement is divided into five stages, as shown in Figure 2. Among these, the length measurement process consists of three main steps. The first step involves detecting the endpoints of the body parts to be measured from photos taken from both the front and side views. The second step calculates the circumference between the detected endpoints, and the third step estimates the total circumference of the desired body part by integrating the results from the front and side views. In this section, we explain the detailed methodology and implementation process for each step and discuss the technical improvements introduced to overcome existing limitations.

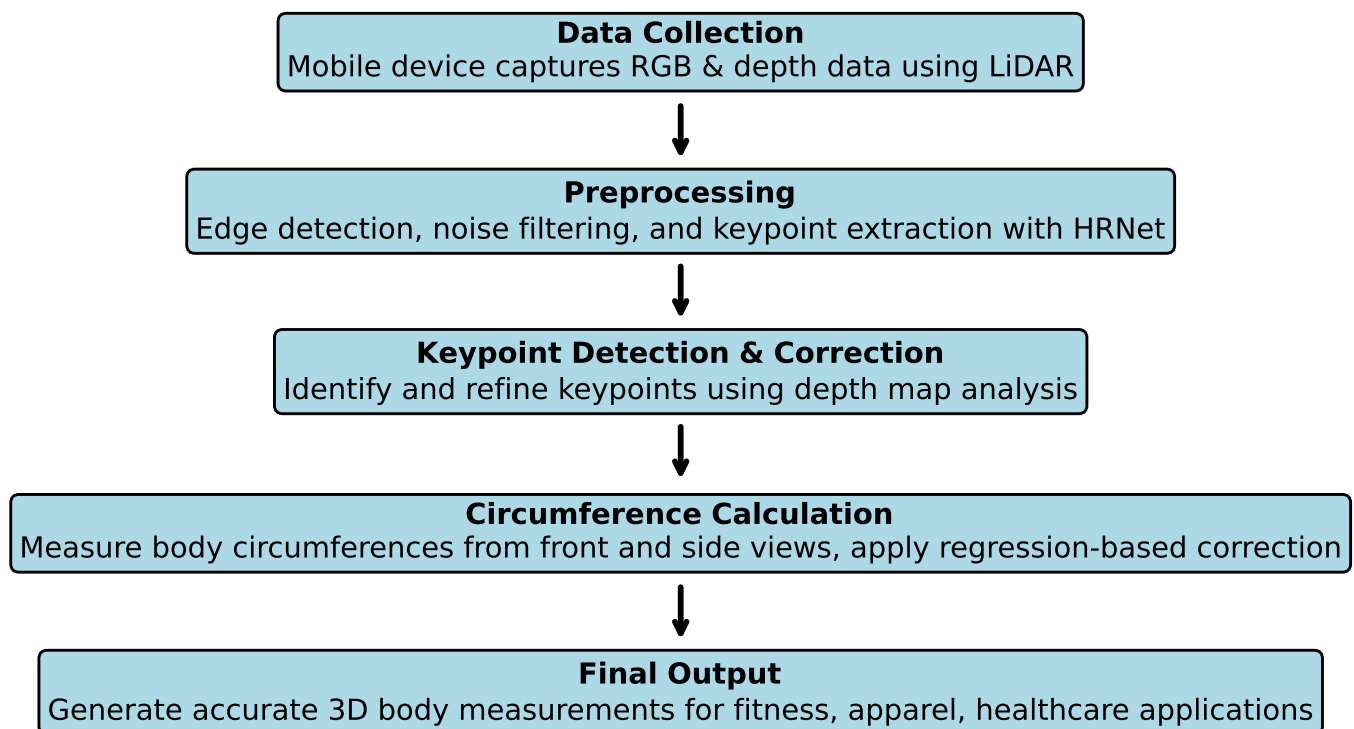


Figure 2. Flowchart of 3D Measurement Methodology.

In the preprocessing stage as shown in Figure 2, noise filtering and edge detection ensure the clarity of depth information before keypoint extraction. HRNet-based keypoint detection is then applied to accurately locate body keypoints, followed by depth map correction to refine the measurement accuracy.

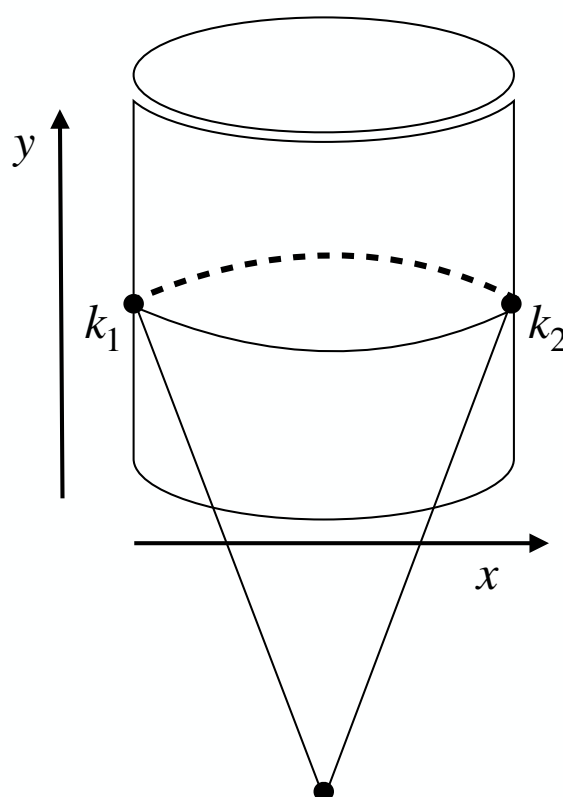
First, in the initial stage of body circumference measurement, accurately detecting the endpoints of body parts is essential. To achieve this, we utilized the Human Pose Estimation using the High-Resolution Representations (HRNet) model [12]. HRNet excels

at maintaining high-resolution representations while precisely detecting keypoints of the body, allowing for accurate identification of various body parts' locations. However, the basic HRNet model provides only a limited number of keypoints necessary for standard pose estimation, which restricts its ability to sufficiently cover all the endpoints required for body circumference measurement.

To overcome these limitations, we applied transfer learning in this study. Specifically, we labeled 1000 images of people by categorizing them into two groups: upper body and lower body. In the upper body category, we labeled the endpoints of the chest and waist, and in the lower body category, we labeled the hip and buttock as keypoints. By further training the HRNet model using this labeled dataset, we aimed to expand the model's limited keypoint detection capabilities. The transfer learning process involved collecting 1000 images of people divided into upper and lower body categories, manually labeling the keypoints for the chest, waist, buttock, hips, and femoral condyles in each image, and setting up based on the pre-trained HRNet model. We then retrained the last few layers of the HRNet model using the labeled dataset to accurately detect the new keypoints, evaluated the model's performance with a validation dataset, and adjusted hyperparameters as necessary.

In the second stage, we calculate the circumference between the detected endpoints from the front and side views. This calculation is primarily based on the Euclidean distance. Specifically, we define two keypoints as $k_1 = (x_1, y_1)$ and $k_2 = (x_2, y_2)$, and represent the distance between the two points as $d(k_1, k_2)$. The Euclidean distance formula is as follows Equation (1) depending on the Figure 3:

$$d(k_1, k_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (1)$$



Camera Position

Figure 3. Circumference estimation using keypoints k_1, k_2 detected from the camera's perspective.

Under the assumption that the y-axis positions of the two points are identical, the straight-line distances in the x-axis direction are summed to approximate the circumference between the two points. Represents this calculation follows Equation (2):

$$\begin{aligned} \text{circum}(x_1, x_2) = & \\ & d(x_1, x_1 + 1) + \\ & d(x_1 + 1, x_1 + 2) + \\ & d(x_1 + 2, x_1 + 3) + \dots + \\ & d(x_2 - 1, x_2) \end{aligned} \quad (2)$$

However, the keypoints detected in 2D images do not correspond to their actual positions in 3D space. Therefore, calculating the total circumference by simply summing the front and back circumferences introduces errors. This is because the boundary surface observed from the camera's viewpoint (2D keypoint) does not coincide with the actual boundary surface (3D keypoint) assumed to represent the object's true contour.

- l : the circumference error caused by the discrepancy between the 2D keypoint detected in the image and the actual 3D keypoint.
- r : the radius of the circle (object).
- θ : the angle between the line connecting the 2D keypoints (detected in the image) and the line connecting the center of the circle (object) from the camera's perspective.
- d : the distance between the camera and the center of the circle (object).

In this study, we simplify the waist-measurement process, using a smartphone LiDAR sensor, to a top-down perspective as illustrated in the figure above. The circle in the figure is assumed to be the waist as viewed from above. Given the characteristics of the LiDAR sensor and the camera, it is almost impossible to clearly capture direct 3D keypoints (i.e., points on the actual perimeter of the waist), and as a result, there is inevitably a discrepancy between the circumference estimated from 2D keypoints and the actual 3D circumference. As shown in Figure 4, let l be the distance between the actual 3D keypoint and the 2D keypoint; even if only one side is considered, the estimated circumference can differ by at least l . In other words, if one simply calculates the front and back circumferences from the 2D keypoints and sums them, the result will necessarily deviate from the actual circumference. Consequently, the discrepancy l in the circumference on one side, caused by the difference between the actual and 2D keypoints, can be derived by the Equation (3):

$$\begin{aligned} \sin\theta &= \frac{r}{d} \\ \theta &= \arcsin\frac{r}{d} \\ l &= r\theta = r \times \arcsin\frac{r}{d} \end{aligned} \quad (3)$$

Therefore, the discrepancy between the 2D keypoints and their 3D counterparts can be represented as $r \times \arcsin(\frac{r}{d})$. Simply adding the circumference of the front portion to that of the back portion in order to estimate the total circumference can lead to inaccuracies. Furthermore, to ensure that the detected keypoints consistently capture the same positions, this study introduces a depth map-based correction procedure. We used a depth map generated by photographing the same scene with an iPhone 12 Pro LiDAR sensor, where lower values indicate proximity to the object and higher values indicate greater distance. Applying the Canny edge algorithm allowed us to detect abrupt depth variations, enabling the adjustment of 2D keypoints to better align with actual object boundaries.

Therefore, the discrepancy between the 2D keypoints and their actual 3D counterparts can be expressed as $r \times \arcsin(\frac{r}{d})$. Since the 2D keypoints detected from a single viewpoint do not correspond to their actual positions in 3D space, simply summing the front circumference (C_f) and back circumference (C_b) to estimate the total circumference (C_t) can introduce errors. This discrepancy arises because 2D keypoints are projected coordinates rather than actual 3D surface points and are affected by occlusion. To address this issue, this study employs regression analysis to estimate the total circumference (C_t) by treating the front circumference (C_f) and side circumference (C_s) as variables. This approach is necessary because projection mismatch prevents accurate 3D measurements using only 2D keypoints. Additionally, to improve the accuracy of 2D keypoint detection, we applied a depth map-based correction technique. This process refines the alignment of the detected 2D keypoints with actual object boundaries, thereby enhancing the accuracy of circumference measurement for various body regions.

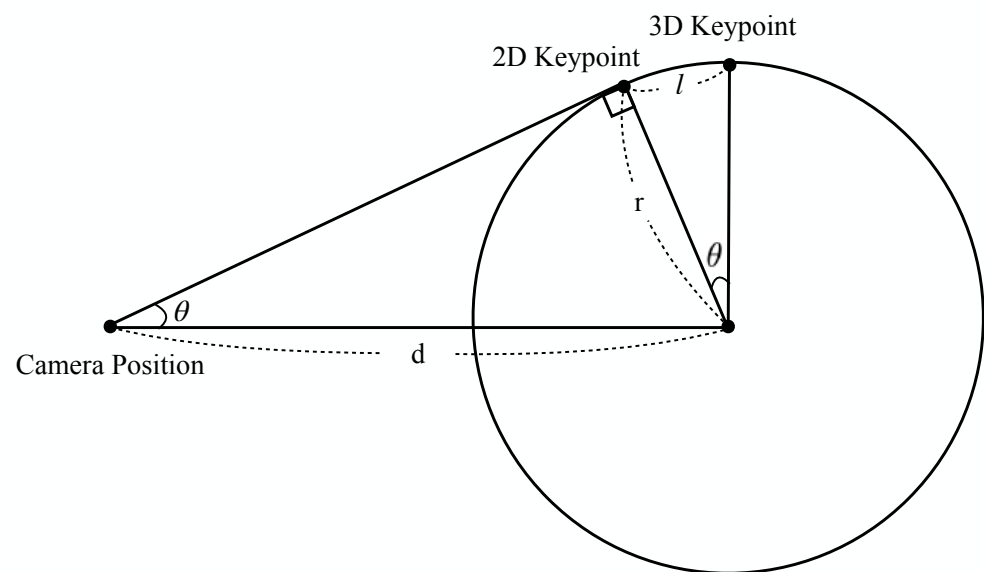


Figure 4. This shows the geometric relationship between 2D and 3D keypoints for the top view of an object in circumference estimation.

This process consists of two main steps. First, the RGB image resolution is scaled to match that of the depth map. For example, a LiDAR-based depth map corresponding to the same scene as the RGB image is obtained, and the Canny edge algorithm is applied to detect boundary surfaces where the depth values exhibit steep gradients [24]. Next, the keypoint coordinates detected in the 2D RGB image are mapped to the depth map coordinate system (i.e., scaled accordingly). Each keypoint is then repositioned to the nearest edge point in the vicinity of the transformed coordinate. By doing so, the keypoints consistently reflect the object's boundary across various conditions such as changes in viewing angle or illumination. Consequently, this procedure provides a reliable set of reference points for measuring both the front and side circumferences, thereby helping to reduce errors.

First, a depth map of the same scene as the RGB image captured using the iPhone's LiDAR sensor is generated. A depth map indicates that lower values are closer to the object and higher values are further away from the object. Next, the Canny edge algorithm is applied to the depth map to detect points where the depth value changes abruptly. This is useful for accurately identifying the object's boundary surfaces. Then, the coordinates of the keypoints detected in the 2D RGB image are transformed to match the scale of the depth map and moved to the nearest edge point. This process includes the following two steps: matching the resolution of the RGB image with that of the depth map.

To illustrate the proposed approach, we used an RGB image with a resolution of 1440×1920 and a depth map with a resolution of 192×256 . First, the keypoint coordinates detected in the RGB image were scaled by a factor of 7.5 to map them onto the depth map coordinate system. Next, each keypoint was repositioned to the nearest edge point identified in the depth map. Through this correction procedure, the 2D-detected keypoints more reliably align with boundary surfaces even under diverse conditions, thereby substantially improving the accuracy of circumference measurements.

Using these corrected keypoints, both the front circumference and the side circumference were calculated. The front circumference was derived from keypoints located on both sides of the object in a front-view image, whereas the side circumference was measured using keypoints from an image in which the object was rotated by 90 degrees. By combining these two measurements, the total circumference was estimated via regression analysis. Based on ground-truth circumference measurements collected from 150 Korean male participants, an optimal regression constant was then determined to enhance the accuracy of the estimated total circumference.

The total circumference (C_t) is calculated using the following regression model on the Equation (4):

$$C_t = \beta(C_f + C_s) + \epsilon \quad (4)$$

Here,

- C_t : Total body circumference (actual value, dependent variable)
- C_f : Front circumference (independent variable)
- C_s : Side circumference (independent variable)
- β : Regression coefficient (derived constant)
- ϵ : Error term

Initially, a method was used in which separate constants were applied to the front (C_f) and side (C_s) circumferences individually. However, when the actual waist circumference was used to classify participants as lean (72 cm or less) or obese (95 cm or more) and then to compare measurement results, it was observed that both the average error and the coefficient of determination (R^2) were adversely affected for lean and obese body types. In reality, C_f and C_s are not independent but rather parts of the same body, inherently influencing each other. Therefore, a method was chosen in which a single constant is multiplied by the sum of C_f and C_s . This approach yielded a lower mean absolute error (MAE) and a higher coefficient of determination. Based on these findings, the total circumference was ultimately optimized by simply summing C_f and C_s before multiplying by a constant.

According to Table 1, the summation-then-constant method demonstrated superior accuracy, reducing MAE from 6.8 cm to 5.2 cm while improving the coefficient of determination R^2 from 0.76 to 0.84. This result suggests that combining front and side circumferences before applying a single correction factor provides a more robust estimation approach.

Waist Circumference: (5)

$$(\text{Front Circumference} + \text{Side Circumference}) \times 1.32$$

Chest Circumference:

$$(\text{Front Circumference} + \text{Side Circumference}) \times 1.427$$

Hip Circumference:

$$(\text{Front Circumference} + \text{Side Circumference}) \times 1.322$$

Buttock Circumference:

$$(\text{Front Circumference} + \text{Side Circumference}) \times 1.53$$

In this study, the derived regression constant exhibited consistently high performance, irrespective of body type (e.g., lean or obese). This result indicates that the combination of front and side circumferences effectively captures an individual's body shape and may maintain a certain level of measurement accuracy across diverse ethnicities and nationalities. Nevertheless, to thoroughly validate this finding, future research should establish a large-scale dataset encompassing a broad range of ethnic and national groups to evaluate the generalizability of the regression constant.

Table 1. Comparison of the separate-constant method and the summation-then-constant method.

Method	Mean Absolute Error (MAE, cm)	Coefficient of Determination (R^2)
Separate-constant method ($\alpha_1 C_f + \alpha_2 C_s$)	6.8	0.76
Summation-then-constant method ($\beta(C_f + C_s)$)	5.2	0.84

3.2. Ablation Study: Performance Comparison of Each Component

In this study, to verify how each component of the proposed model affects body circumference measurement performance, we compared the following three settings:

- Model A: Keypoint detection (only)
- Model B: Keypoint detection + LiDAR sensor
- Model C: Keypoint detection + correction algorithm + LiDAR sensor (proposed model)

Table 2 summarizes the relative error (%) and runtime (seconds) for each setting.

Table 2. Comparison of body circumference measurement performance for each setting.

Setting	Keypoint Detection	LiDAR Sensor	Correction Algorithm	Relative Error (%)
Model A (1)	Yes	No	No	Not measurable
Model B (2)	Yes	Yes	No	19.2
Model C (3, Ours)	Yes	Yes	Yes	5.2

Model A (keypoint detection only): Since it relies solely on a single 2D image (keypoints), it is impossible to measure actual length due to the absence of depth information, leading to unmeasurable results.

Model B (keypoint + LiDAR): By including LiDAR sensor data, it becomes possible to measure actual dimensions, unlike Model A. However, since the measurement process uses only keypoints detected from 2D RGB images, the relative error reached 19.2%.

Model C (keypoint + correction algorithm + LiDAR, proposed model): By adding a correction algorithm on top of keypoints and LiDAR sensor data, the relative error was reduced to 5.8%. This indicates that the depth-map-based correction algorithm significantly contributes to improving the accuracy of body circumference measurements.

Consequently, the proposed Model C, which employs a correction algorithm to more precisely utilize the 3D information from LiDAR, achieves both accuracy and practicality. This ablation study provides a quantitative verification of how each component such as keypoint detection, LiDAR sensor, and correction algorithm, contributes to performance improvement.

3.3. Utilization of Uniform 3D Data

In this study, to enhance the accuracy and consistency of the body measurement algorithm, experiments were conducted using uniform 3D data such as cylinder and mannequin. These standardized 3D objects, which closely resemble the human body while exhibiting minimal deformation, were useful for systematically verifying the performance of the algorithm in its initial stages. Cylinders, with their simple structure, are ideal objects for evaluating whether the algorithm can accurately measure complex body structures.

Additionally, mannequin, which mimic the shape of the human body, were used to test the measurement accuracy of various body parts. Through this approach, it was confirmed that the algorithm could accurately measure body parts of different shapes and sizes. The experimental procedure involved first capturing RGB images and depth maps of the cylinder and mannequin from various angles using an iPhone's LiDAR sensor. Using an HRNet-based keypoint detection model, the endpoints of the major parts of each object were detected. The detected keypoints were then adjusted to their accurate positions in 3D space through the aforementioned correction process. Based on the corrected keypoints, circumferences were calculated, and the total circumference was estimated by integrating the measurements taken from the front and side views. The measured circumference values were compared with the actual values to evaluate the accuracy of the algorithm. Through these experiments, the accuracy and consistency of the proposed body circumference measurement algorithm were systematically validated, and methods to minimize errors in measuring various body parts were established.

The body circumference measurement algorithm proposed in this study consists of the following key components. First, RGB images and depth maps are extracted at the same scale to facilitate subsequent data merging and analysis. Second, the keypoints detected by the HRNet model are adjusted to accurate 3D positions through a depth map-based correction process. Third, the circumferences of body parts are calculated using the corrected keypoints, and the total circumference is estimated by integrating the measurements taken from the front and side views. Finally, a regression equation was derived to estimate the total circumference based on the measured circumferences from the front and side views, enabling accurate measurements. Additionally, to enhance the efficiency of the algorithm, the following optimization techniques were introduced. First, model lightweighting and parallel processing were employed to enable real-time body circumference measurements. Data captured from various angles and lighting conditions were utilized to improve the model's generalization performance. Furthermore, filtering techniques were applied to minimize noise in the depth maps.

3.4. Experimental Environment and Dataset

As shown in Figures 5 and 6, uniform 3D data such as cylinder and mannequin were used to ensure the consistency and reproducibility of the experiments. In Figures 5 and 6, (b) in each figure is a depth map visualizing the distance between the camera and the object. Lower depth values are represented by darker colors, while higher depth values are represented by brighter colors, visually depicting the three-dimensional structure of the object. The actual circumference of the cylinder is 17.27 cm, and the estimated circumference measured by LiDAR is 17.68 cm, demonstrating an accuracy within a 3% error margin. The results for the cylinder and mannequin are shown in Table 3, respectively. The regression equation is tailored to humans, and the Mannequin has a different shape compared to an actual person, resulting in slightly larger errors. The experimental environment involved using a model equipped with an iPhone's LiDAR sensor to simultaneously collect RGB images and depth maps, maintaining data consistency by photographing objects under the same lighting conditions and fixed distances. Through such standardized datasets, the performance of the algorithm could be systematically evaluated, and errors that could occur in measuring various body parts could be analyzed.

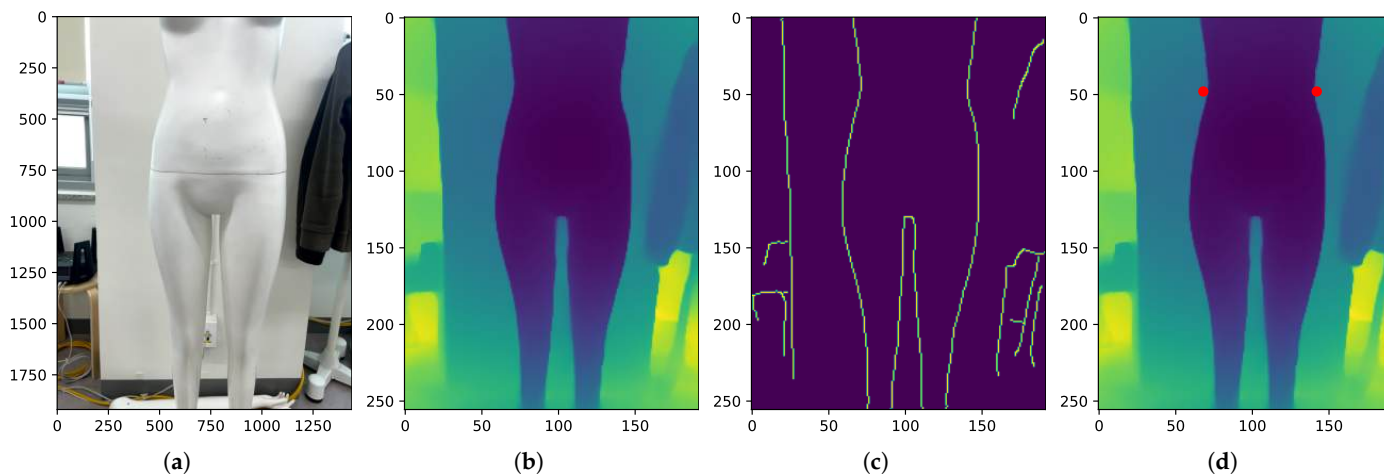


Figure 5. The (a) shows the mannequin's RGB photo, (b) is depth map, (c) is edge detection using the Canny algorithm, and (d) is keypoint detection in the depth map photo.

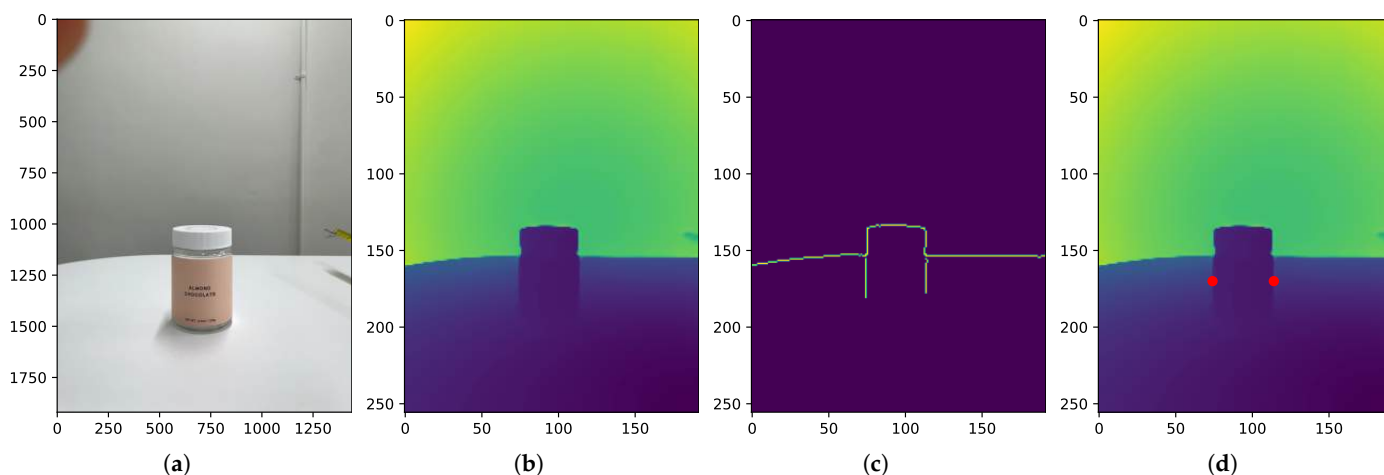


Figure 6. The (a) shows Cylinder's RGB photo, (b) is depth map, (c) is edge detection using the Canny algorithm, and (d) is keypoint detection in the depth map photo.

Table 3. Body measurements of the mannequin by body part. The sizes obtained from the LiDAR sensor are compared with the actual sizes.

		Actual (cm)	Estimated (cm)	Error (%)
Mannequin	Waist	55	46.7	18
	Chest	60	61.2	2
	Hip	62	57.19	5
	Buttock	63	67.18	1.08
Cylinder		17.27	17.68	3

Figure 7a,c visualize the distance measurements before correction for the cylinder and mannequin, respectively. Distance values exceeding 12 cm between adjacent pixels were observed, which can be interpreted as the distance between the object and the background rather than within the object itself. Points where the values suddenly increase indicate locations beyond the object's boundary. In contrast, Figure 7b,d show the distance measurements after correction for the cylinder and mannequin, respectively, with the maximum value not exceeding 1 cm. This suggests that by detecting edges using depth values and correcting the keypoints, the lengths to the object's boundary surfaces were accurately measured.

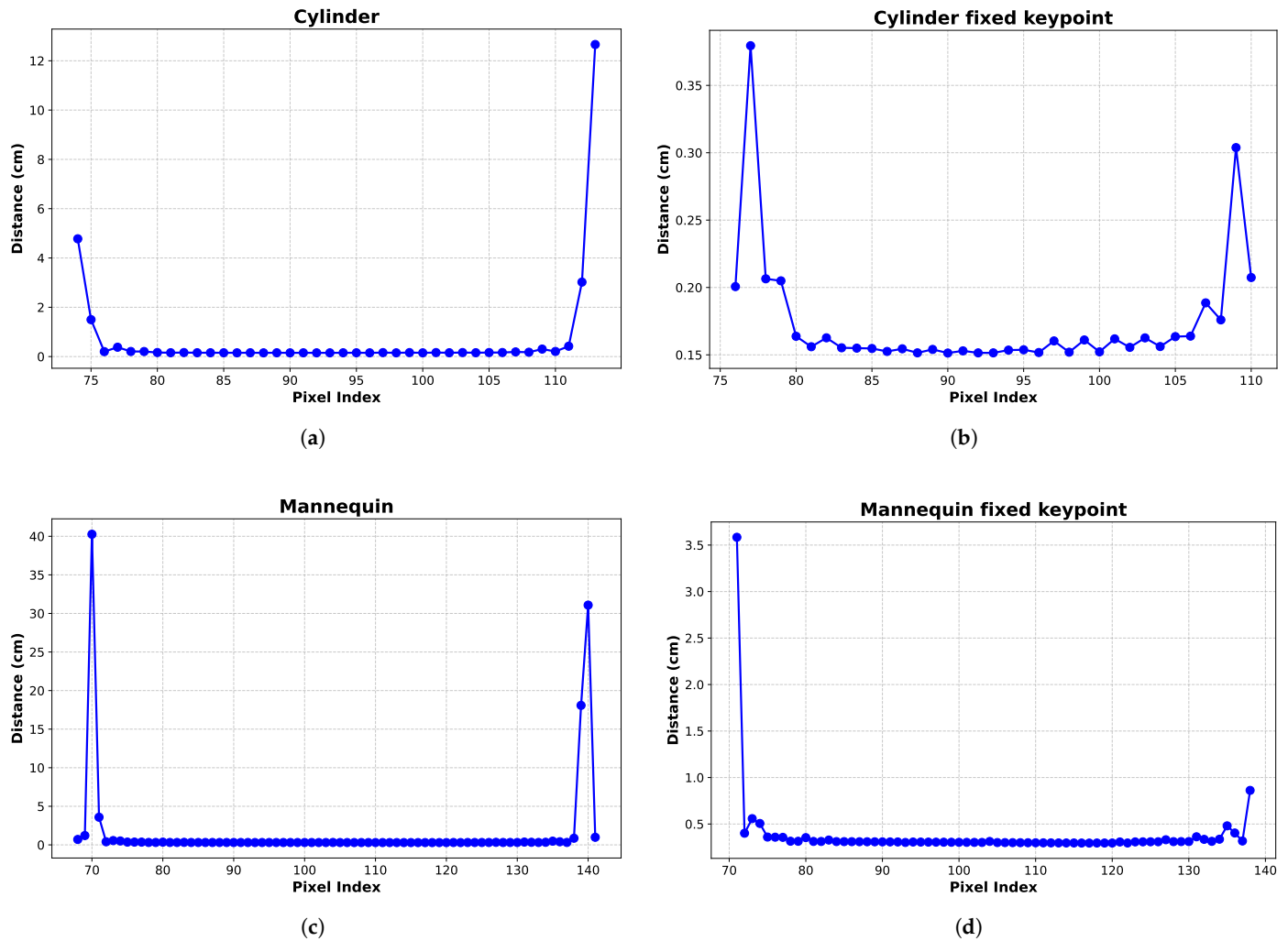


Figure 7. The graphs (a) is display the cylinder’s uncorrected keypoints, (b) is corrected keypoints, as well as (c) is the mannequin chest’s uncorrected keypoints, (d) is corrected keypoints. In the corrected graphs, the y-axis automatically adjusts its maximum value as the pixel distance decreases.

4. Actual Model Measurement

Figure 8 shows photos taken to capture images of the upper and lower body parts of a person, respectively. Due to the characteristics of the iPhone’s LiDAR sensor, the depth map size is 192×256 , so it was extracted from the RGB image at the same scale. This maintains consistency between images, ensuring accuracy in subsequent data merging and correction processes.

Figure 8c shows the edge image obtained by applying the Canny Edge algorithm to the depth image. Canny Edge Detection is an algorithm used to find edges in images, effectively detecting points where depth values change sharply. This algorithm not only identifies points with sudden changes in depth values but also helps to accurately detect the actual boundary surfaces of objects. By first identifying areas with significant depth changes and then connecting points with high depth change intensity, the algorithm can more precisely detect the actual boundary surfaces of the object, rather than meaningless edges. Figure 9 shows arbitrary keypoints manually detected from the RGB image. The process involves manually selecting keypoints first and then automatically correcting them using deep learning. To make the correction effect more apparent, the manually selected keypoints were marked slightly offset from the boundary. The size of the RGB image is 1440×1920 , and the size of the depth map is 192×256 . Since the ratio between the RGB image and the

depth map is 7.5, the keypoint coordinates detected in the RGB image were divided by 7.5 to convert them into depth map coordinates. The transformed keypoints were adjusted to match the resolution of the depth map, which plays an important role in maintaining the accurate alignment between the two datasets.

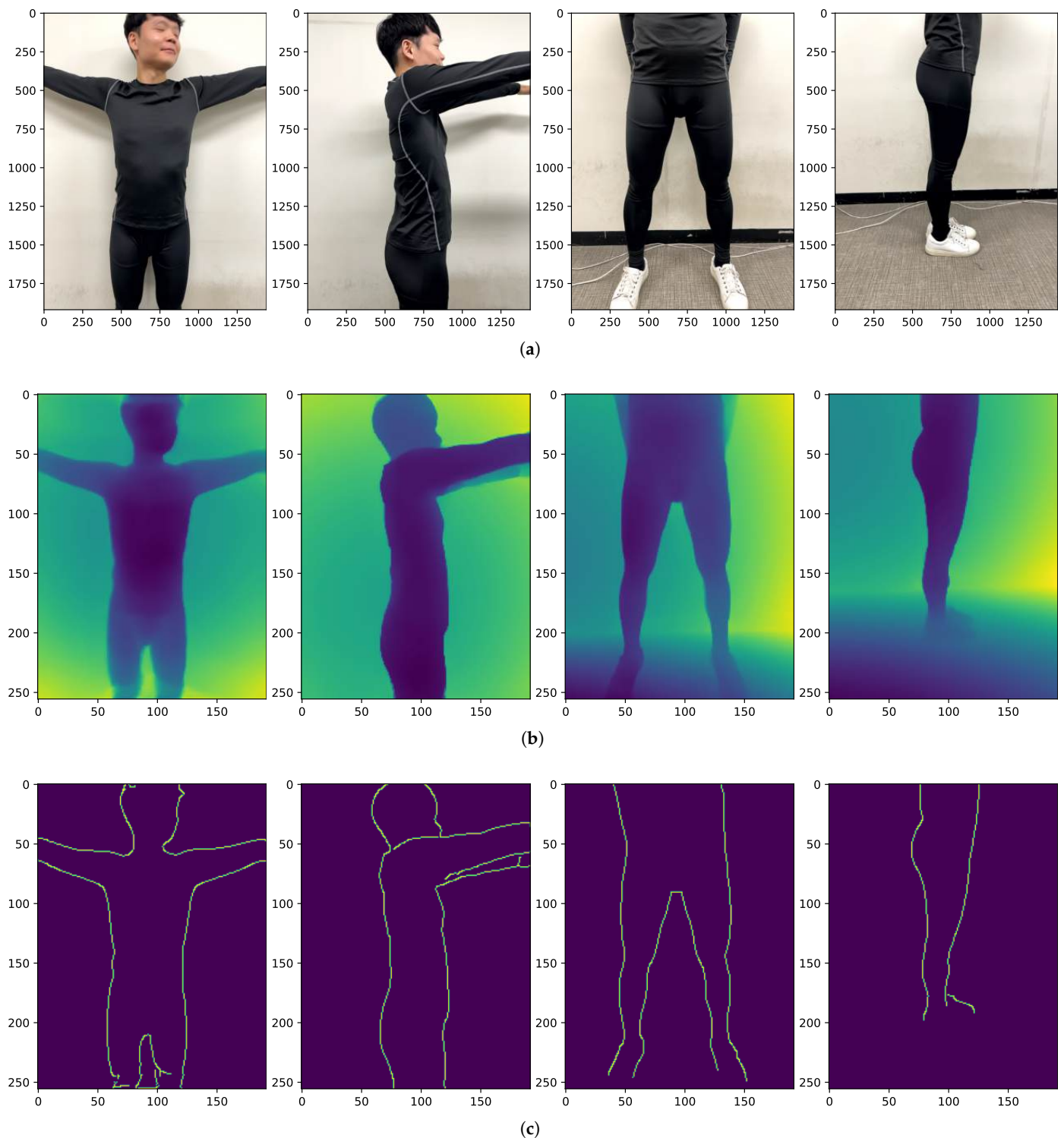


Figure 8. The (a) shows the model's RGB capture, (b) is depth map conversion, and (c) is Canny edge detection.

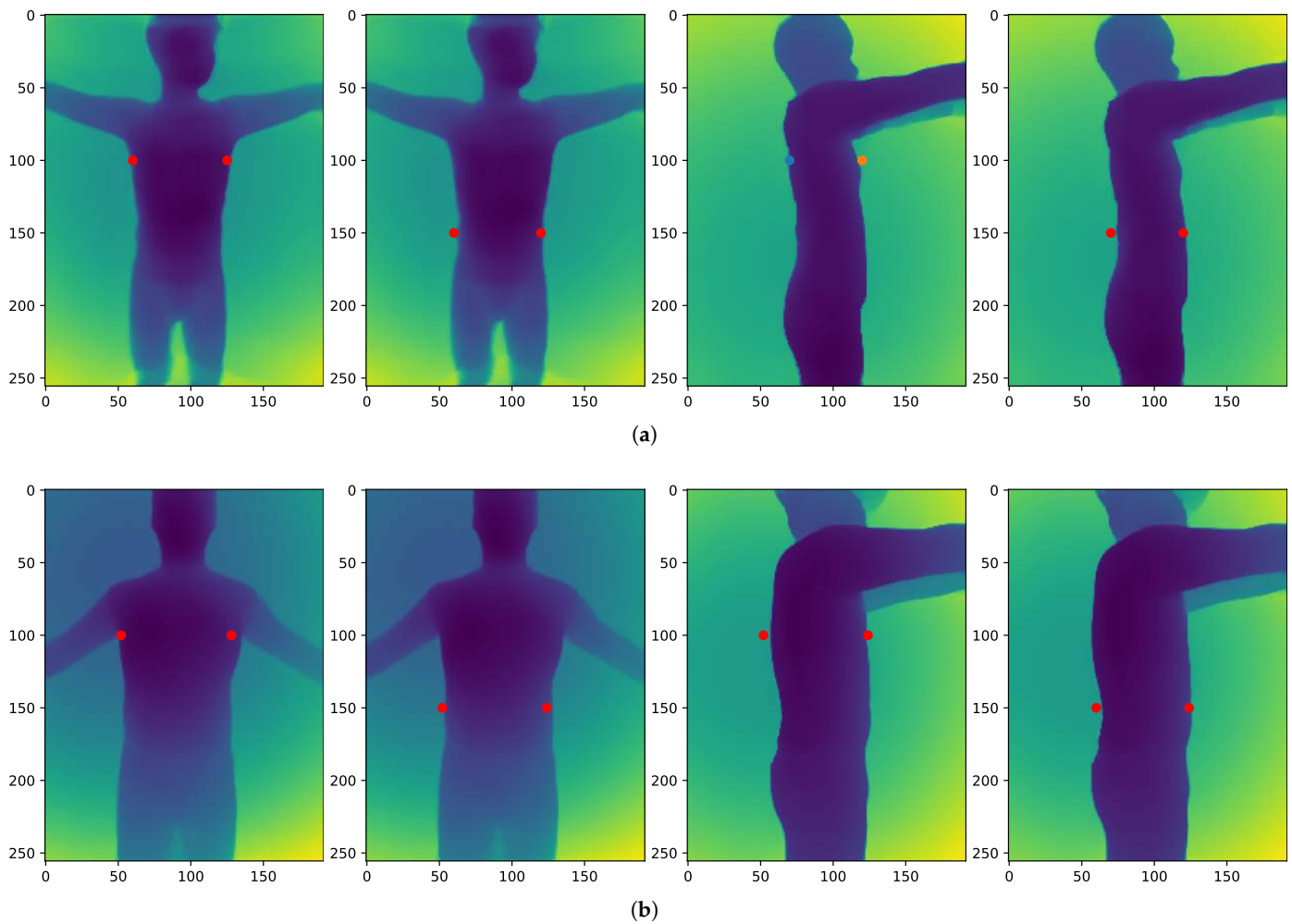


Figure 9. Keypoints were placed at both endpoints of the waist, chest, hips, and buttocks in the front and side views to measure the length of each body part. The (a) shows the Model 1, (b) shows the Model 2. The Model 3 and 4 described in Appendix A.

Figure 10a,b illustrate the keypoint correction process before and after applying the edge algorithm. Figure 10a shows the coordinates of the keypoints manually identified, while Figure 10b presents the results corrected based on the edge algorithm. In Figure 10a,b, the y -axis represents the distance between the subject's pixels and the background, and the x -axis represents the pixel index. In Figure 10, when a keypoint is located outside the subject, the distance between the subject and the wall behind exceeds 10 cm. In contrast, when a keypoint is located inside the subject, it shows a short distance within a few centimeters. The keypoint correction was performed by moving the keypoints to the coordinates of the nearest edges to align them with their actual positions in 3D space. After correction, the keypoints coincide with the edges detected in the depth map, enabling a more accurate representation of the object's actual boundary surfaces.

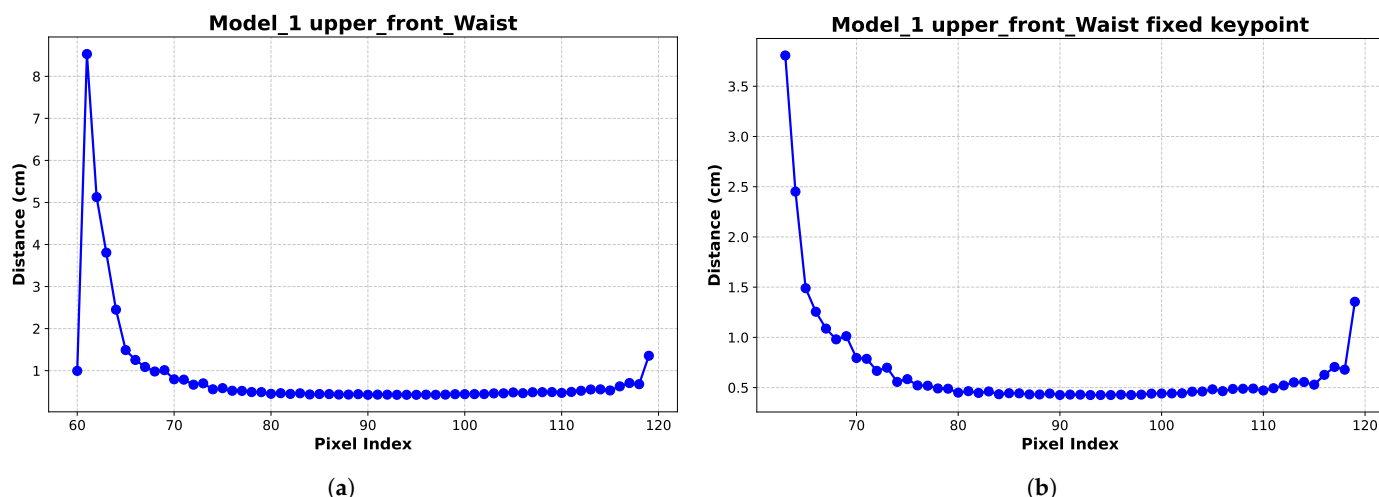


Figure 10. Graphs presenting the data before keypoint correction, where the y-axis represents the distance between the subject's pixels and the background. Graph (a) illustrates Model 1, while (b) illustrates Model 2. The pixel distances for other parts of Model 1 and 2, as well as for Model 3 and 4, are detailed in the Appendix A.

5. Results

To evaluate the performance of the 3D distance measurement algorithm developed in this study, experiments were conducted on four body parts: waist circumference, chest circumference, buttock circumference, and hip circumference. For each body part, the total circumference was estimated based on front and side measurements of a total of four models, and the errors compared to actual measurements were analyzed. Tables 4–7 shows the results comparing the estimated circumferences with the actual measurements for each body part.

Table 4. The results of the waist circumference measurements from the camera's perspective are presented. The total circumference was calculated by multiplying the sum of the front and side lengths by a factor of 1.32.

	Front (cm)	Side (cm)	Estimated (cm)	Actual (cm)	Error (%)
Model 1	34	25	77.98	78	0
Model 2	39.06	30	91.218	95	4.2
Model 3	34	28	81.964	82	0
Model 4	54	43	128.253	121	5.9

Table 5. The results of the chest circumference measurements from the camera's perspective are presented. The total circumference was calculated by multiplying the sum of the front and side lengths by a factor of 1.427.

	Front (cm)	Side (cm)	Estimated (cm)	Actual (cm)	Error (%)
Model 1	34	22	81.2	83	2
Model 2	36	34	101.5	102	4
Model 3	32	30	89.94	90	0
Model 4	50	39	127	121	4.9

Table 6. The results of the buttock circumference measurements from the camera’s perspective are presented. The total circumference was calculated by multiplying the sum of the front and side lengths by a factor of 1.322.

	Front (cm)	Side (cm)	Estimated (cm)	Actual (cm)	Error (%)
Model 1	34	28	82	76	7.8
Model 2	36	30	87	94	7.4
Model 3	34	28	82	82	0
Model 4	51	41	120.32	118	1.96

Table 7. The results of the hip circumference measurements from the camera’s perspective are presented. The total circumference was calculated by multiplying the sum of the front and side lengths by a factor of 1.53.

	Front (cm)	Side (cm)	Estimated (cm)	Actual (cm)	Error (%)
Model 1	35	25	91.8	91	0.8
Model 2	38	30	104	104	0
Model 3	35	26	98	97	1
Model 4	45	32	117.81	121	2.6

Overall, the algorithm recorded relative errors within 8% for most body parts, and particularly showed low error rates below 4% for waist and hip circumferences. This indicates that the algorithm effectively applied keypoint correction and regression analysis methods to enhance the accuracy of body circumference measurements. In measuring the buttock circumference of actual human models, the error rates for Model 1 and Model 2 were relatively high at 7.8% and 7.4%, respectively. This can be attributed to the complex 3D structure of the buttock area and the difficulty in selecting keypoint positions. On the other hand, in measuring hip circumference, all subjects recorded low error rates within 2.6%, confirming the stability of the algorithm.

Additionally, as shown in Figure 10a,b, keypoint correction utilizing the depth map overcame the limitations of 2D image-based measurements and enabled more precise 3D measurements. Experimental results confirmed that the correction process through the depth map contributed to the accuracy improvement for each body part. Moreover, the real-time processing speed in actual experimental environments was sufficiently fast, suggesting practical application possibilities. The overall performance of the algorithm proposed in this study holds significant meaning for the transition from professional equipment to lightweight mobile devices. The conventional 2D length measurement methods were limited in measurement accuracy due to the inability to reflect depth information of the body, but this algorithm overcomes such limitations by effectively utilizing 3D information.

6. Summary and Conclusions

In this study, the proposed measurement algorithm overcame the limitations of conventional 2D-based human measurement techniques by integrating HRNet-based keypoint detection with a depth map-based correction process, thereby improving precision. The main conclusions are as follows:

- **Ensuring Accuracy and Practicality in Body Circumference Measurement:**
Experimental results showed that for most body regions, the relative error remained within 8%, demonstrating higher accuracy compared to existing 2D-based methods. In particular, key areas such as the waist, chest, and hip were measured with consistent performance, confirming the algorithm's potential for practical applications.
- **Proposing a 2D Keypoint Correction Method Using Depth Maps:**
By correcting the 2D keypoints detected in images with the 3D data from depth maps, the influence of factors such as camera angles and lighting variations was mitigated. This enabled more precise measurements in 3D space, complementing the limitations of 2D-based approaches.
- **Achieving Real-Time Capability through Lightweight Processing:**
A simple computational method that calculates only the Euclidean distance after 2D keypoint detection was adopted, reducing the computational burden. As a result, the approach shows the potential for near-real-time operation even in resource-constrained environments such as mobile devices.
- **Potential Applications and Limitations:**
This algorithm is expected to be applicable in diverse fields, including fitness, healthcare, and the apparel industry. Nonetheless, the error was slightly higher for certain body areas, and the experiment was conducted with a limited sample, pointing to existing constraints. Future studies should expand the sample to various nationalities and ethnic groups to enhance the algorithm's generalizability and accuracy.

In conclusion, the 3D landmark-based measurement algorithm proposed in this study achieves both high accuracy and near-real-time performance, offering significantly greater versatility compared to existing methods. Through ongoing refinement and expansion, this algorithm is anticipated to contribute to the development of more precise and reliable body measurement systems. Future work should focus on expanding the dataset to include a wider range of body types and ethnicities to ensure broader applicability. Additionally, further optimization of the algorithm can facilitate seamless integration into mobile applications, enabling real-time 3D body measurement for various practical uses in healthcare, fitness, and apparel industries.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: Data is not publicly available due to privacy or ethical restrictions.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

The complete set of images and graphs for each RGB, Depth, and Canny edge used in the experiment is included.

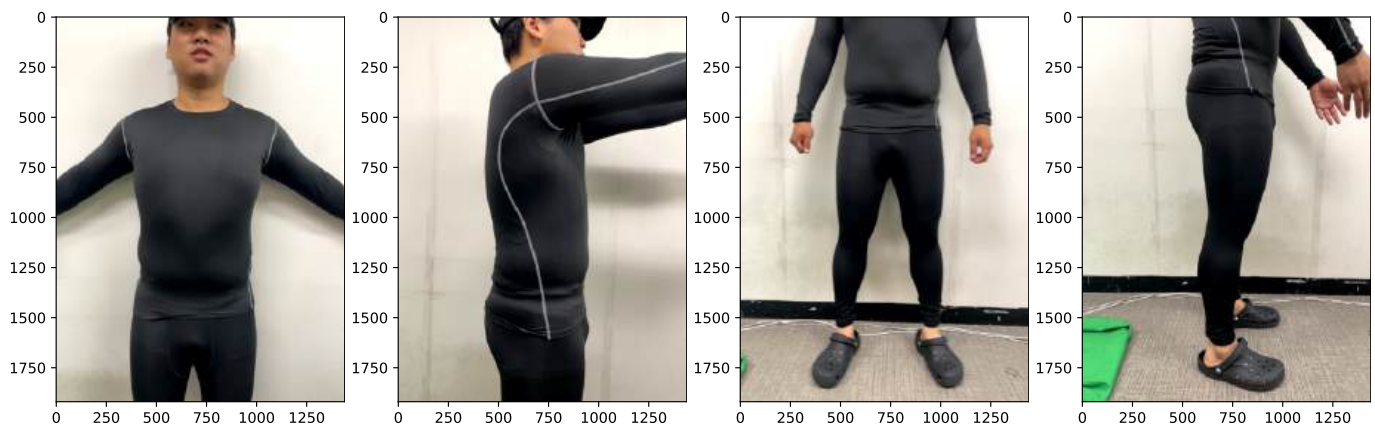


Figure A1. These figures shows the model's RGB capture of Model 2.

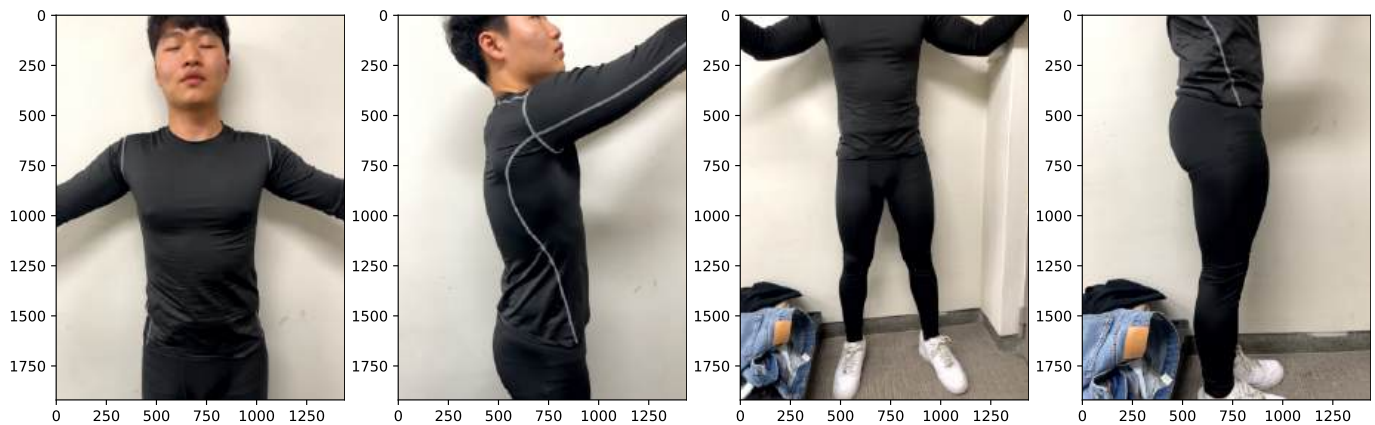


Figure A2. These figures shows the model's RGB capture of Model 3.

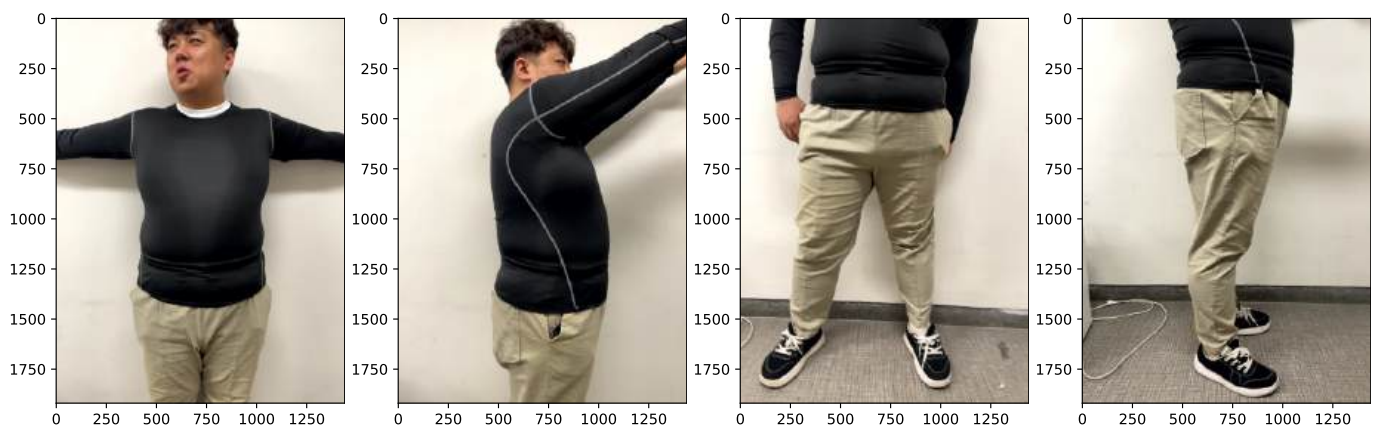


Figure A3. These figures shows the model's RGB capture of Model 4.

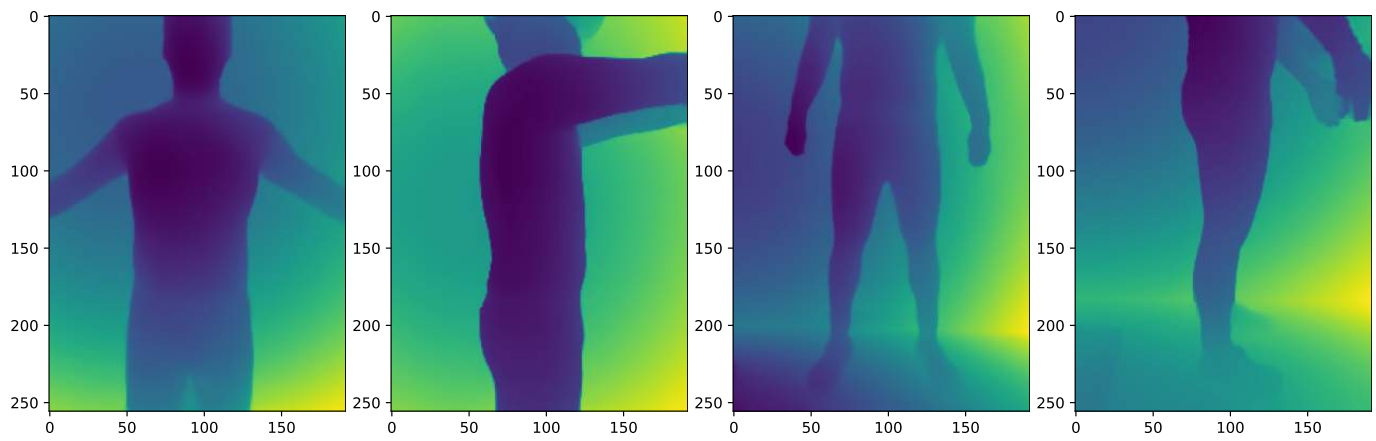


Figure A4. The figure shows the model's depth map conversion of Model 2.

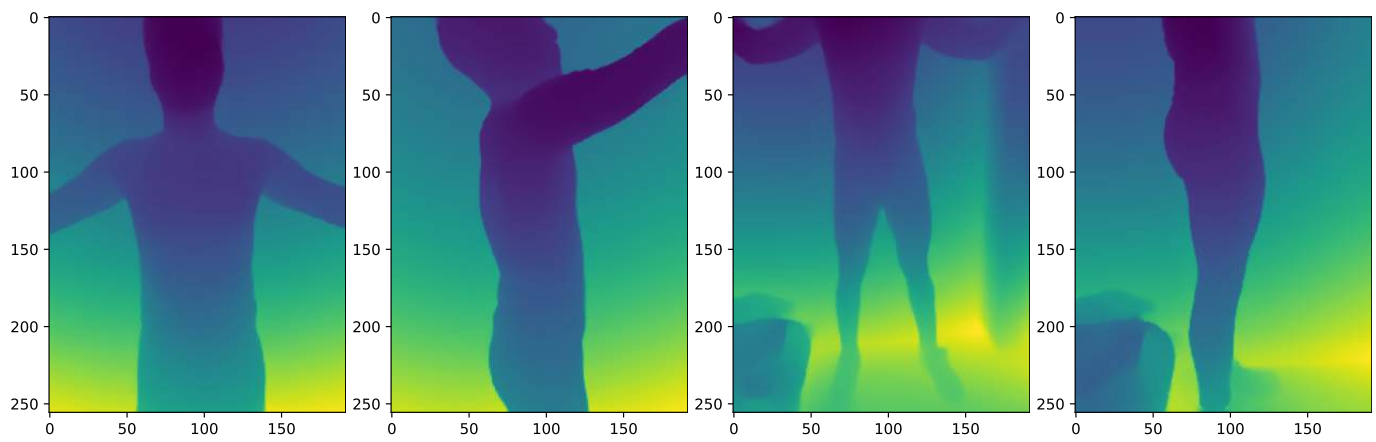


Figure A5. The figure shows the model's depth map conversion of Model 3.

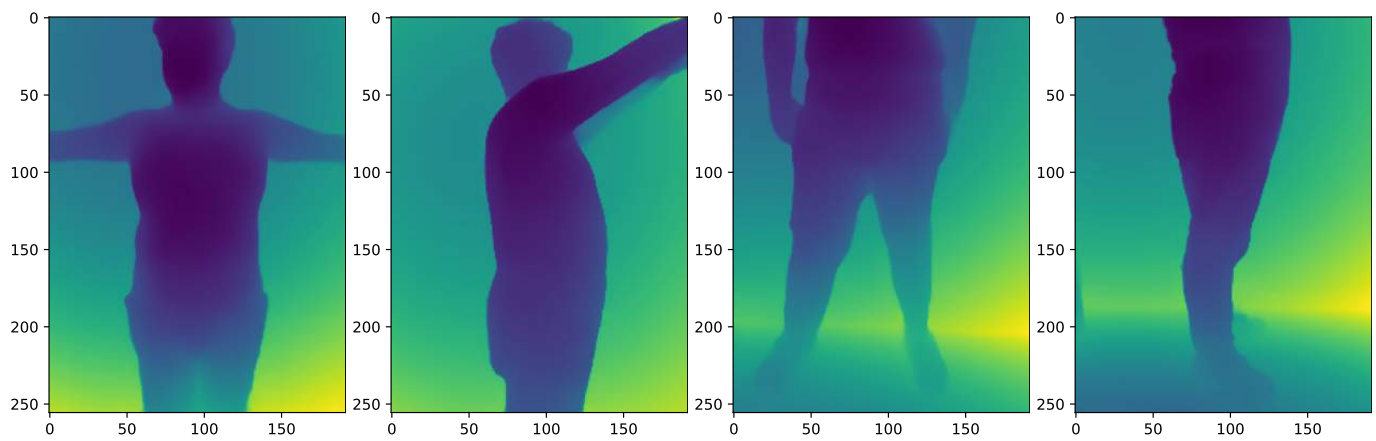


Figure A6. The figure shows the model's depth map conversion of Model 4.

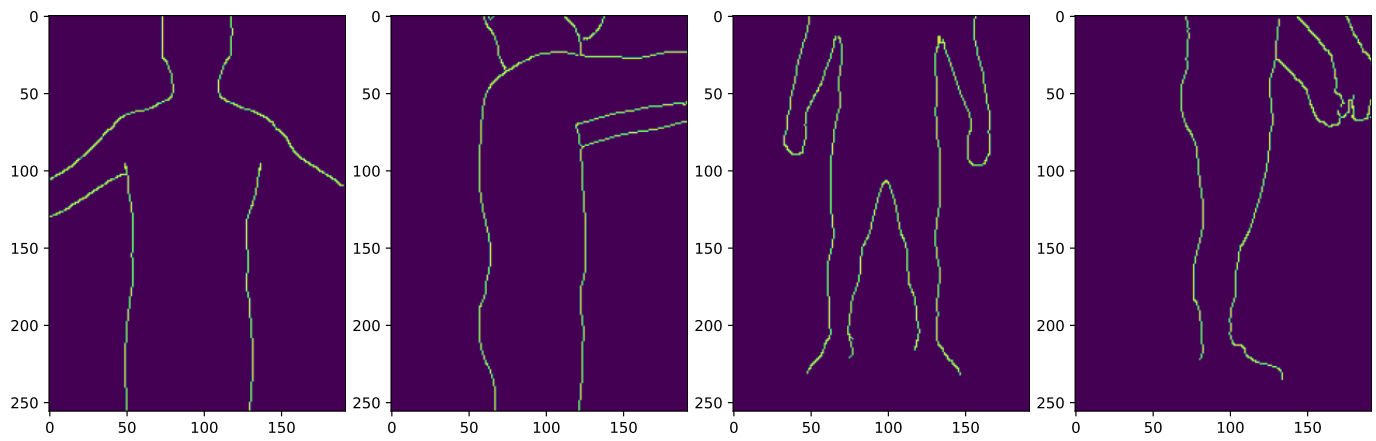


Figure A7. The figure shows the model 2 Canny edge detection.

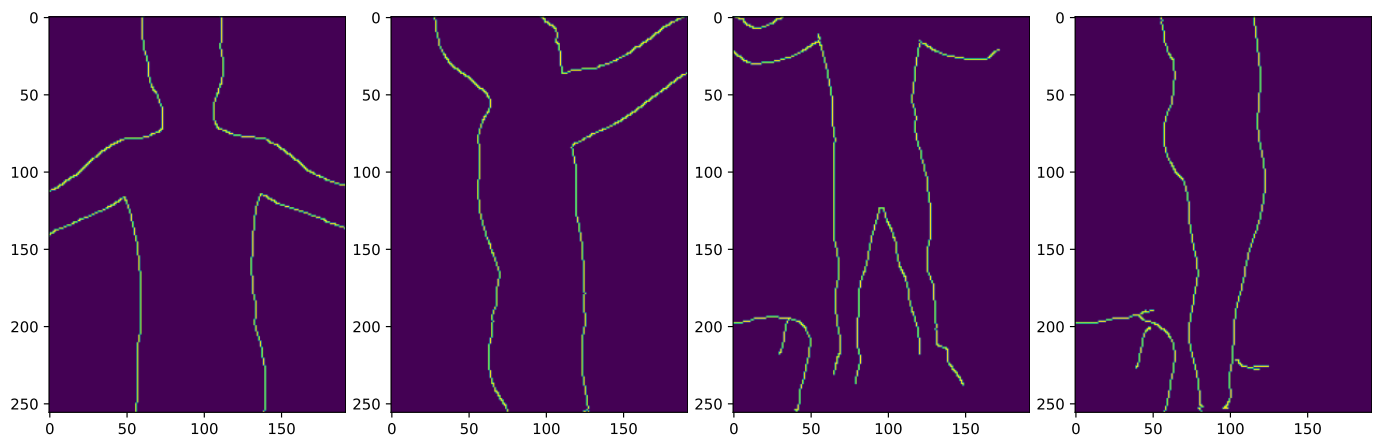


Figure A8. The figure shows the model 3 Canny edge detection.

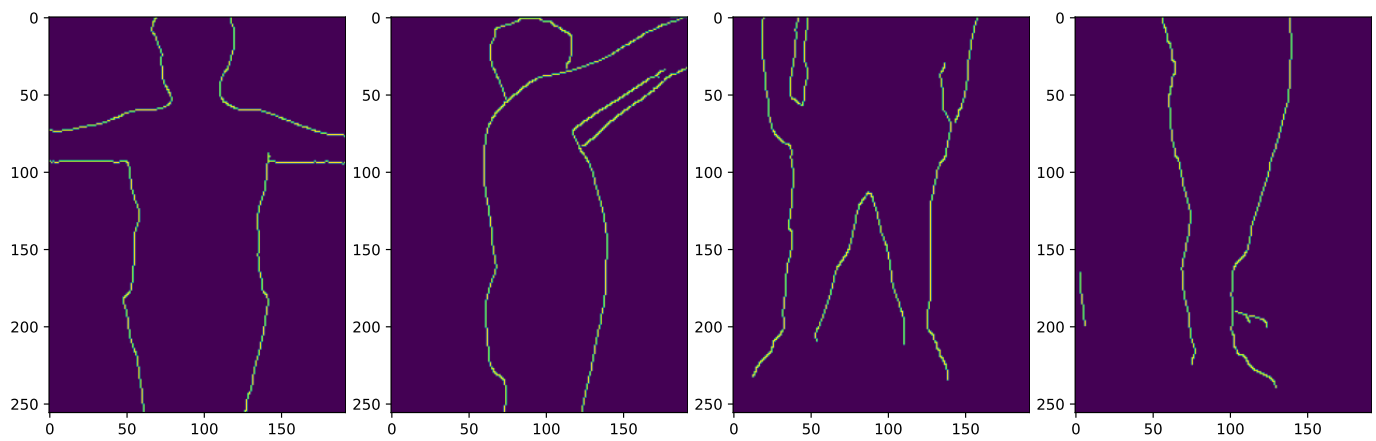


Figure A9. The figure shows the model 4 Canny edge detection.

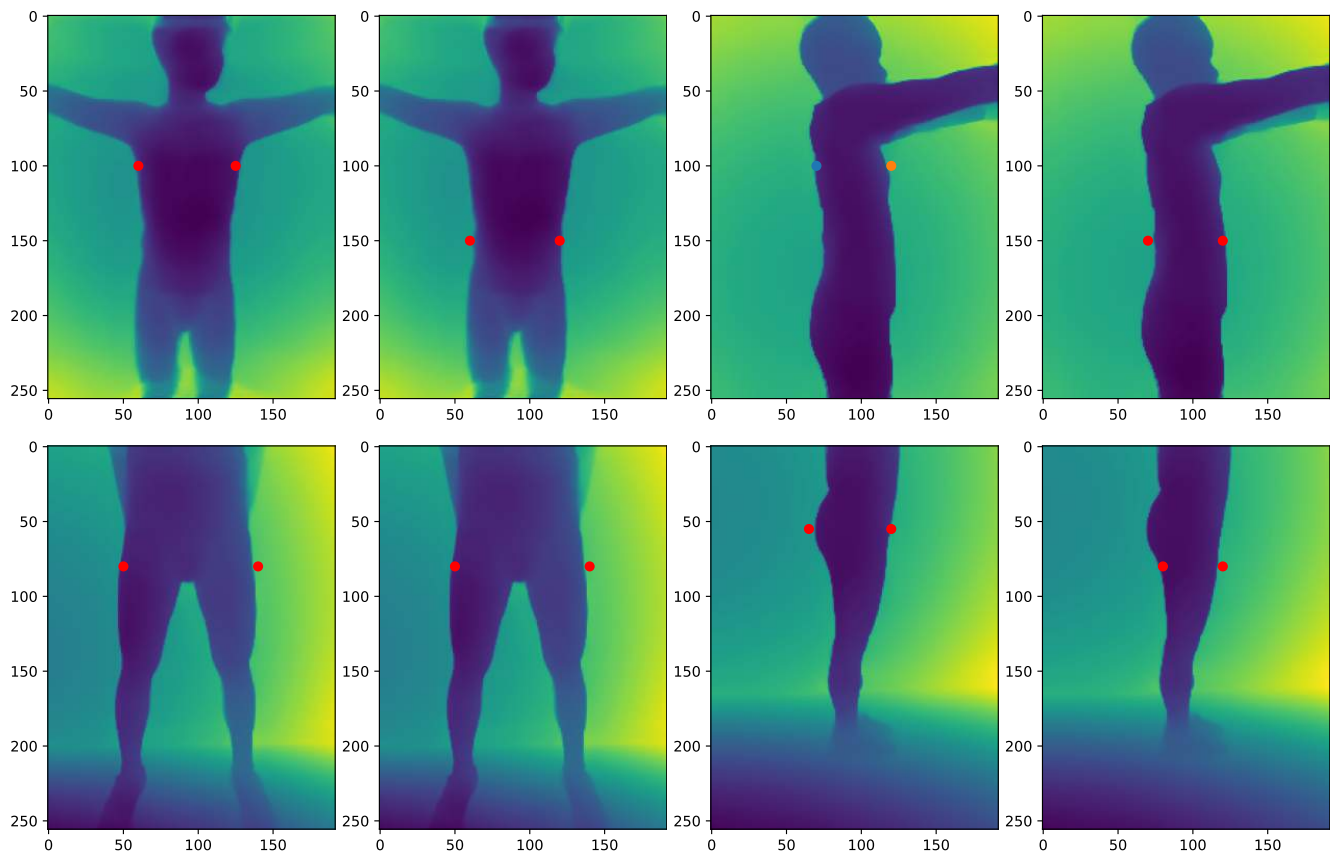


Figure A10. Keypoints were positioned at both ends of the waist, chest, hips, and buttocks in the front and side views of Model 1 to measure each body part's length.

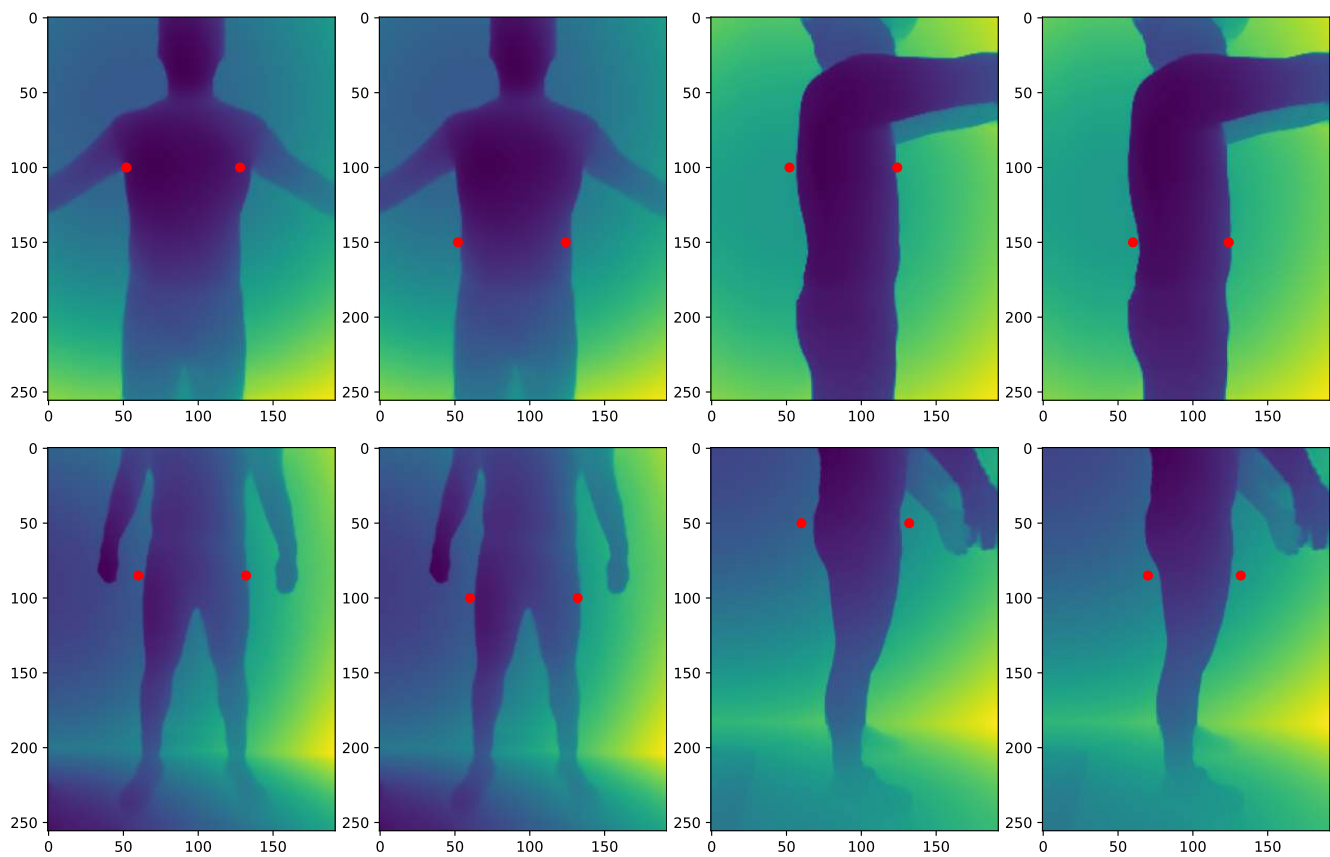


Figure A11. Keypoints were positioned at both ends of the waist, chest, hips, and buttocks in the front and side views of Model 2 to measure each body part's length.

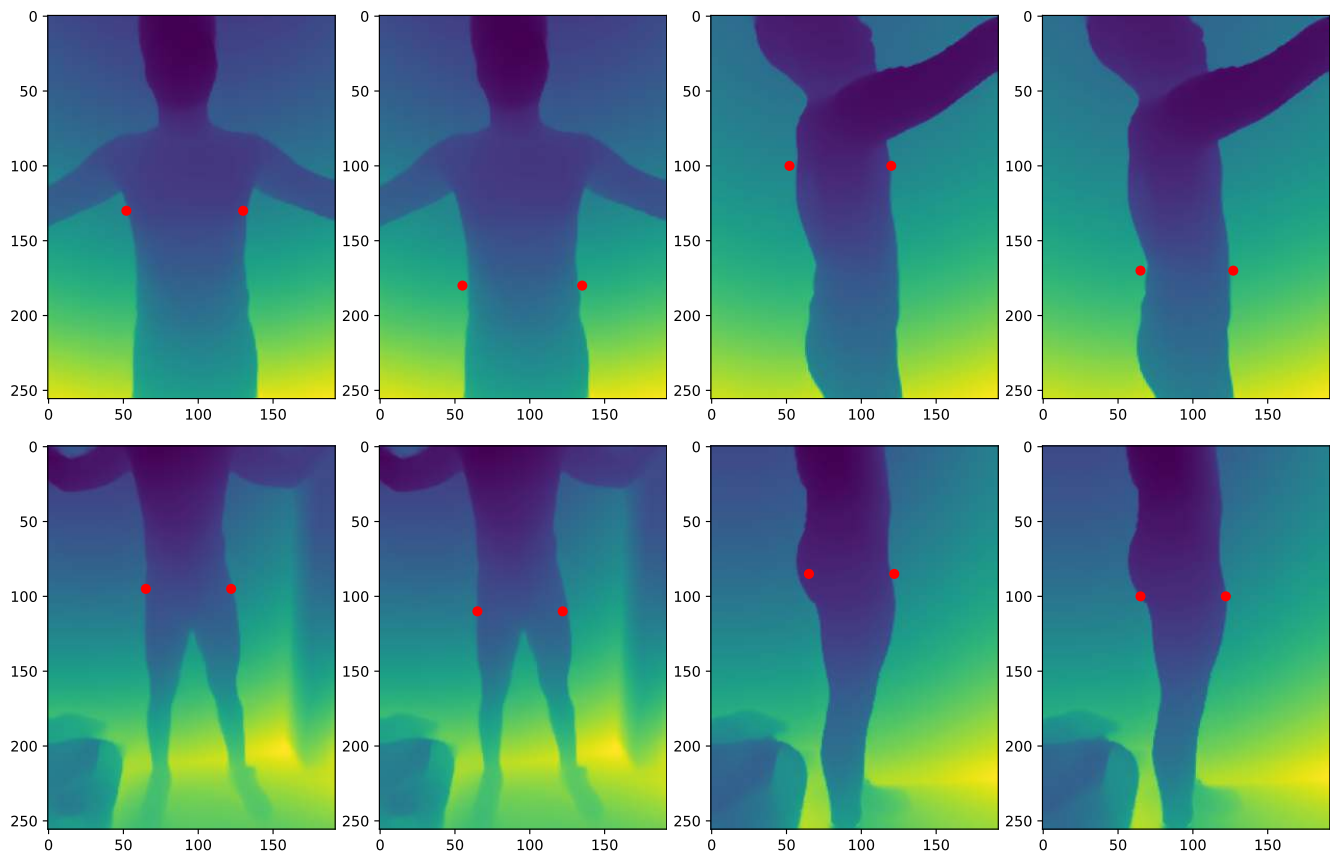


Figure A12. Keypoints were positioned at both ends of the waist, chest, hips, and buttocks in the front and side views of Model 3 to measure each body part's length.

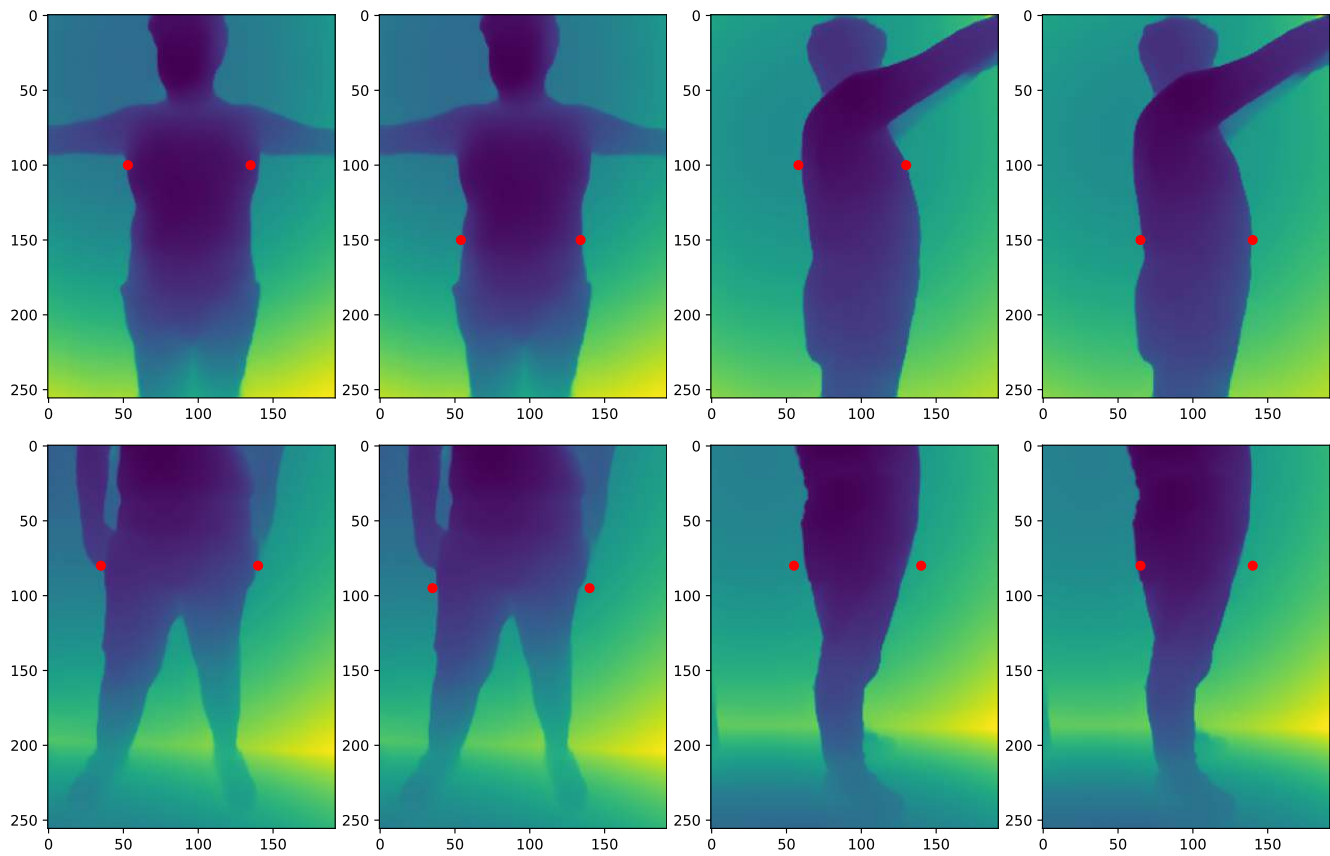


Figure A13. Keypoints were positioned at both ends of the waist, chest, hips, and buttocks in the front and side views of Model 4 to measure each body part's length.

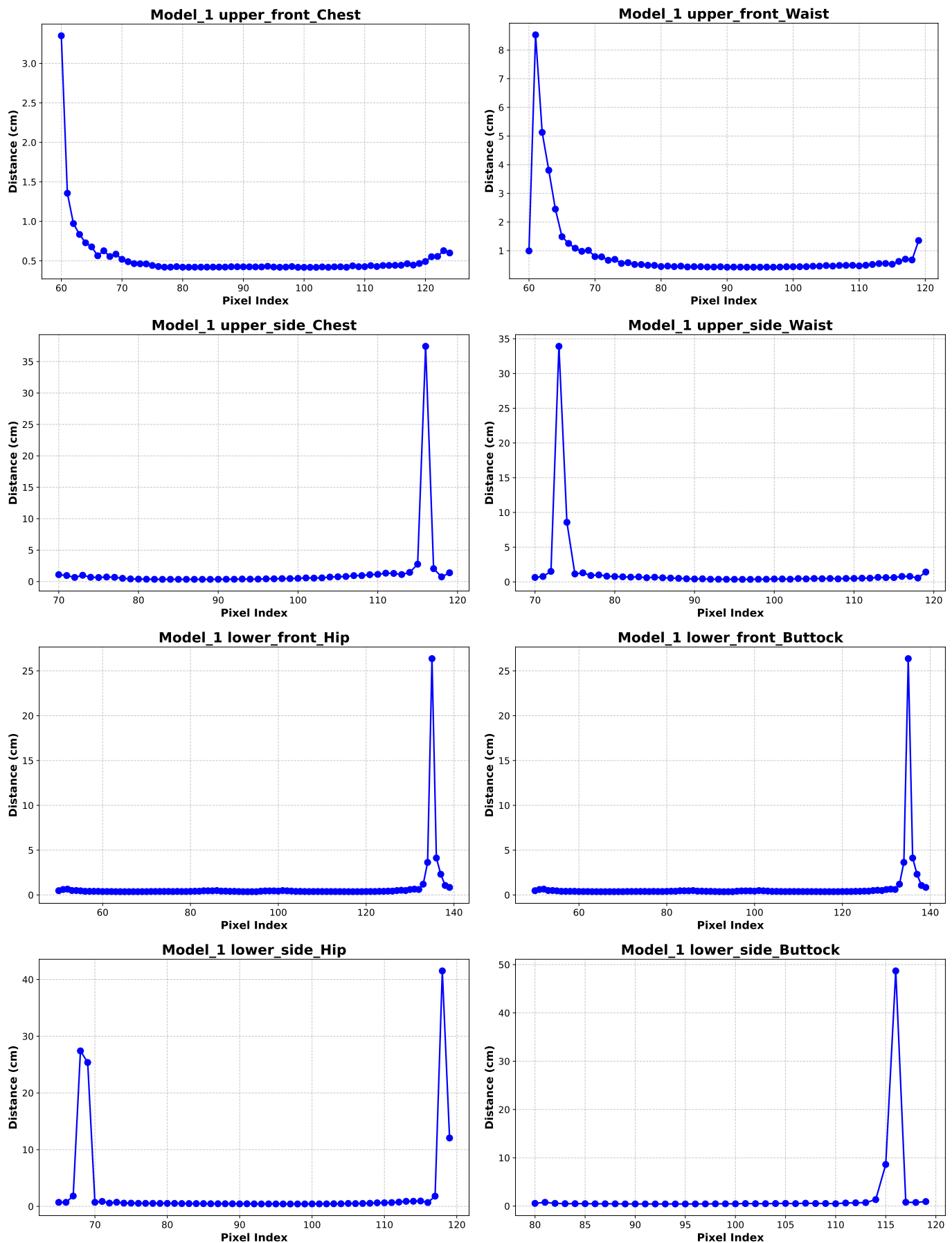


Figure A14. Graphs showing the data before keypoint correction of Model 1, with the y-axis representing the distance between the subject's pixels and the background.

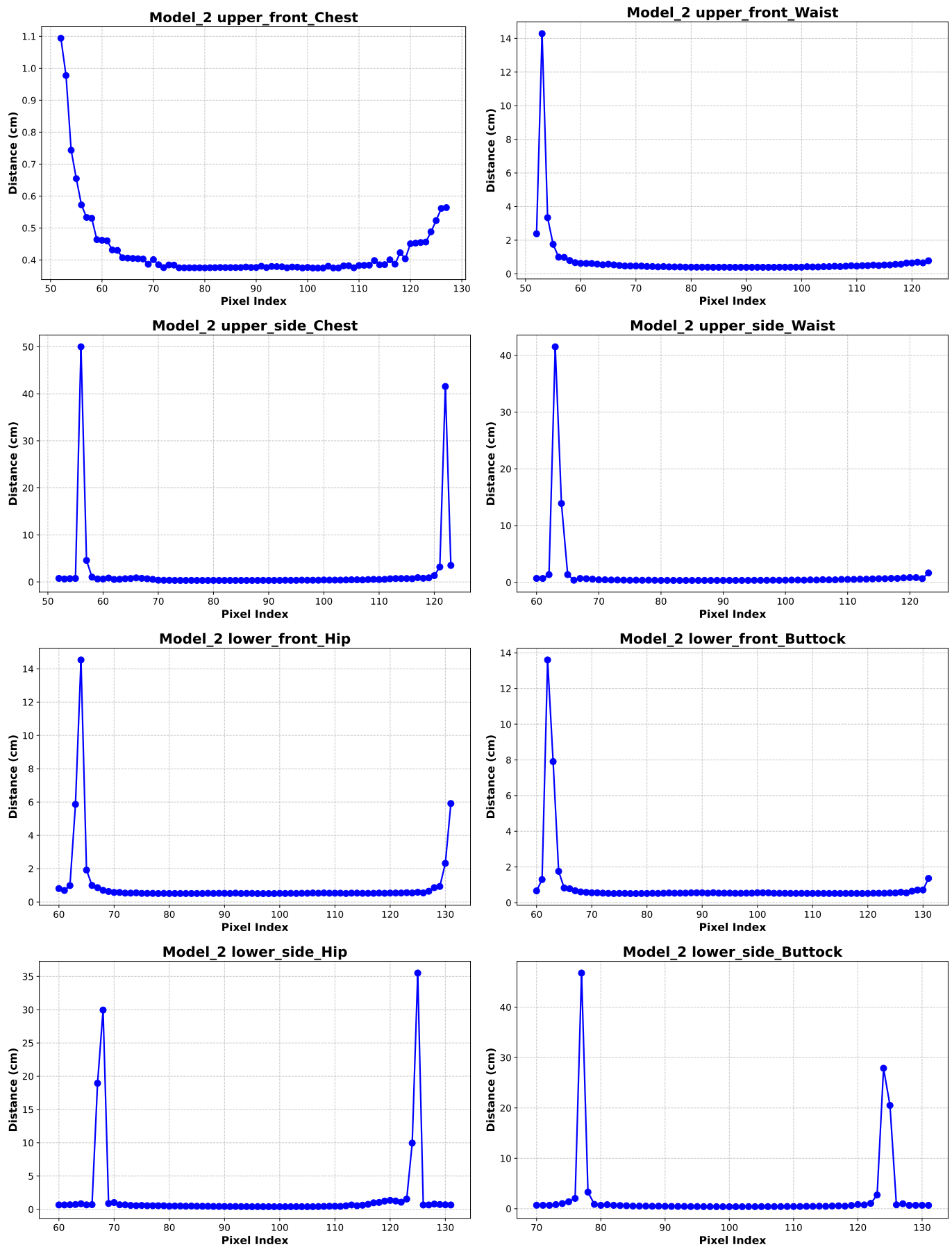


Figure A15. Graphs showing the data before keypoint correction of Model 2, with the y-axis representing the distance between the subject's pixels and the background.

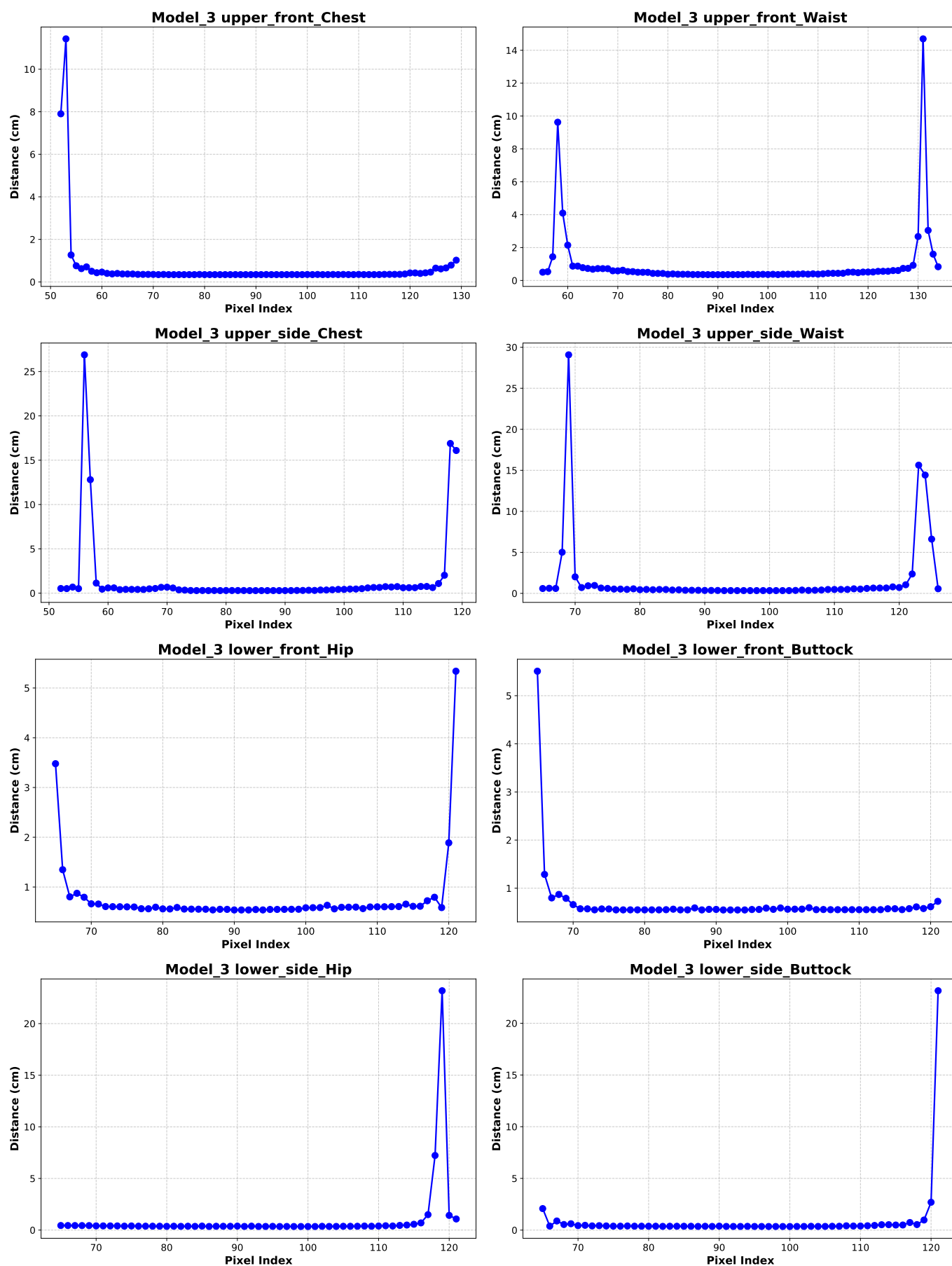


Figure A16. Graphs showing the data before keypoint correction of Model 3, with the y-axis representing the distance between the subject's pixels and the background.

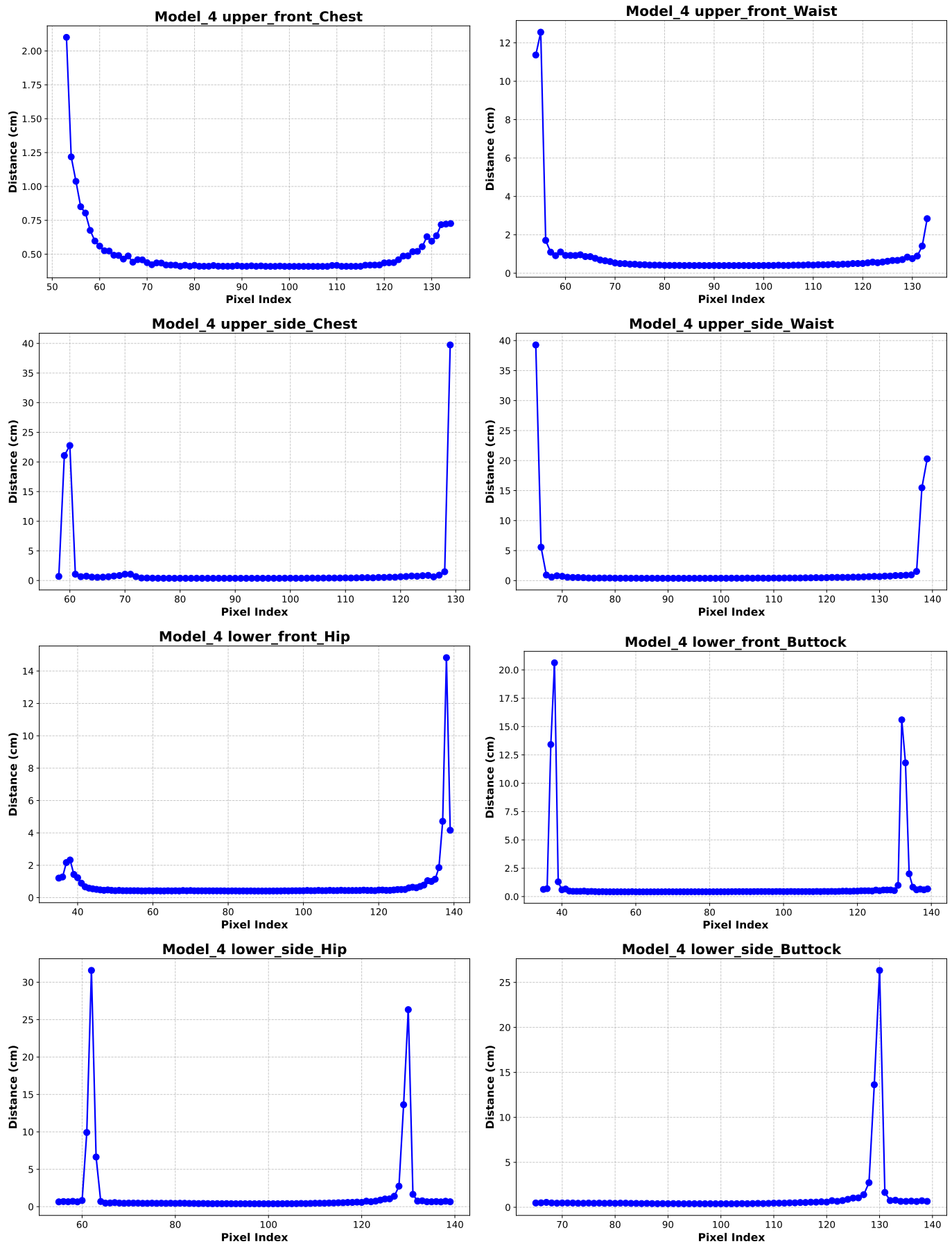


Figure A17. Graphs showing the data before keypoint correction of Model 4, with the y-axis representing the distance between the subject's pixels and the background.

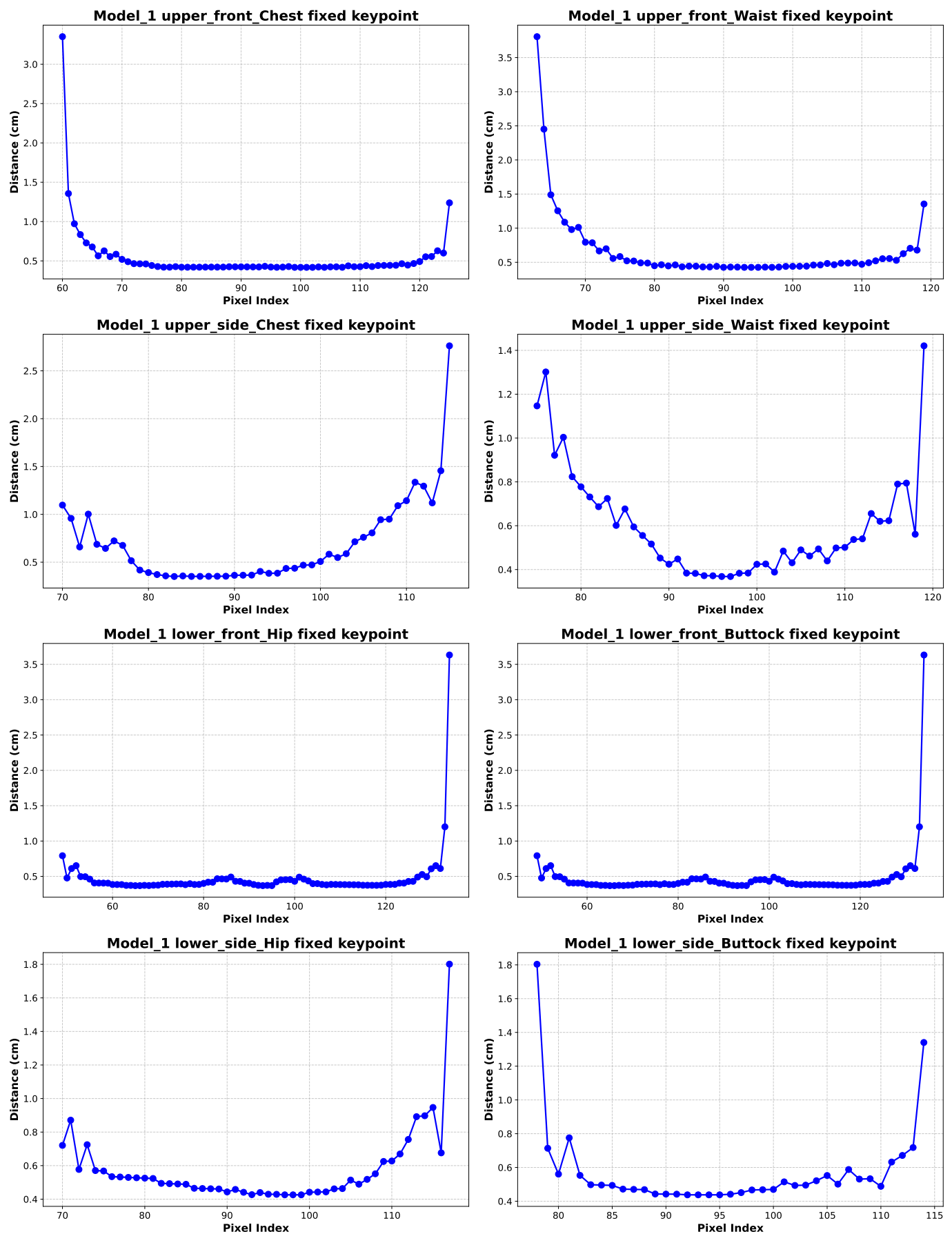


Figure A18. Graphs showing the data after keypoint correction of Model 1. The y-axis automatically adjusts its maximum value as the distance between pixels decreases.

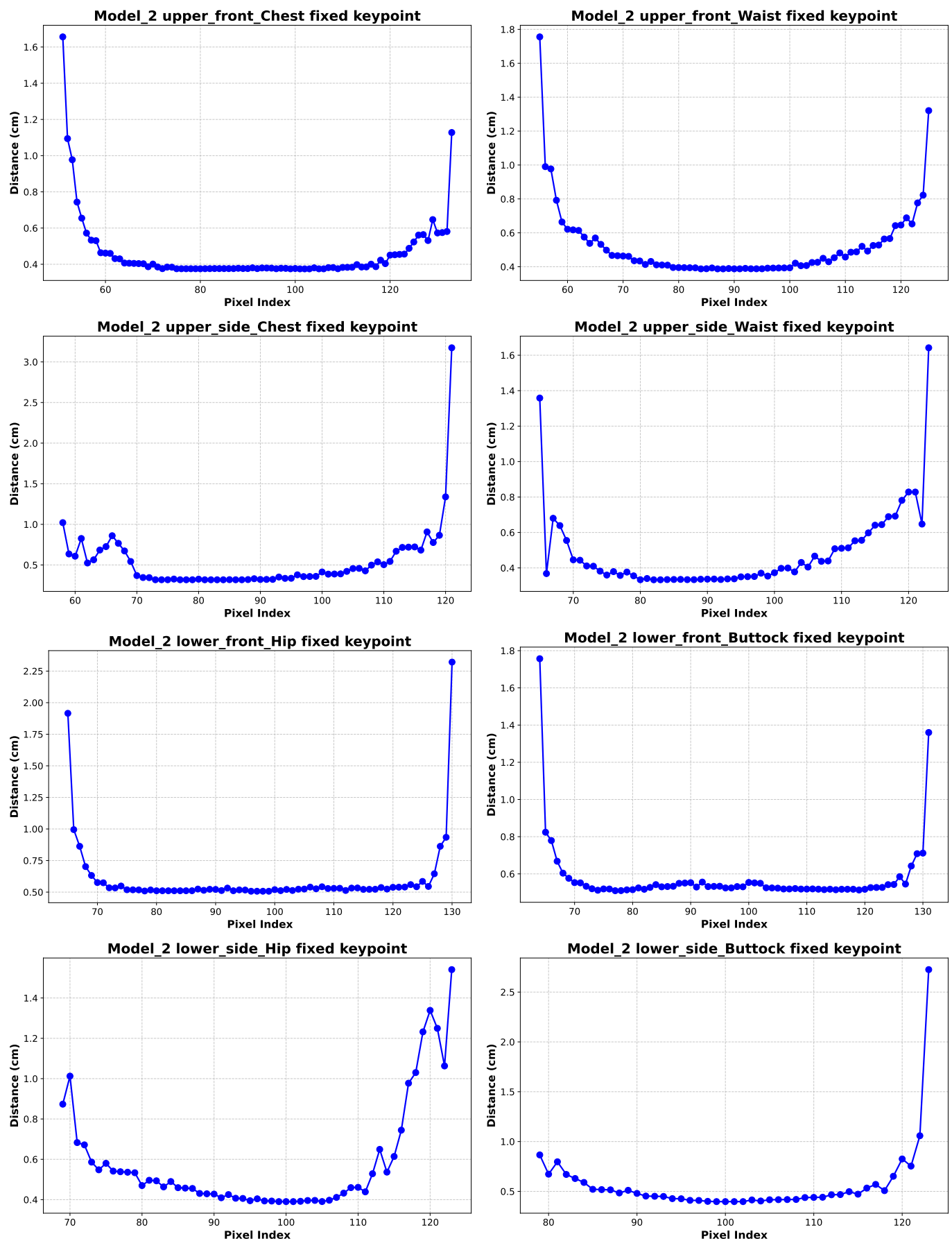


Figure A19. Graphs showing the data after keypoint correction of Model 2. The y-axis automatically adjusts its maximum value as the distance between pixels decreases.

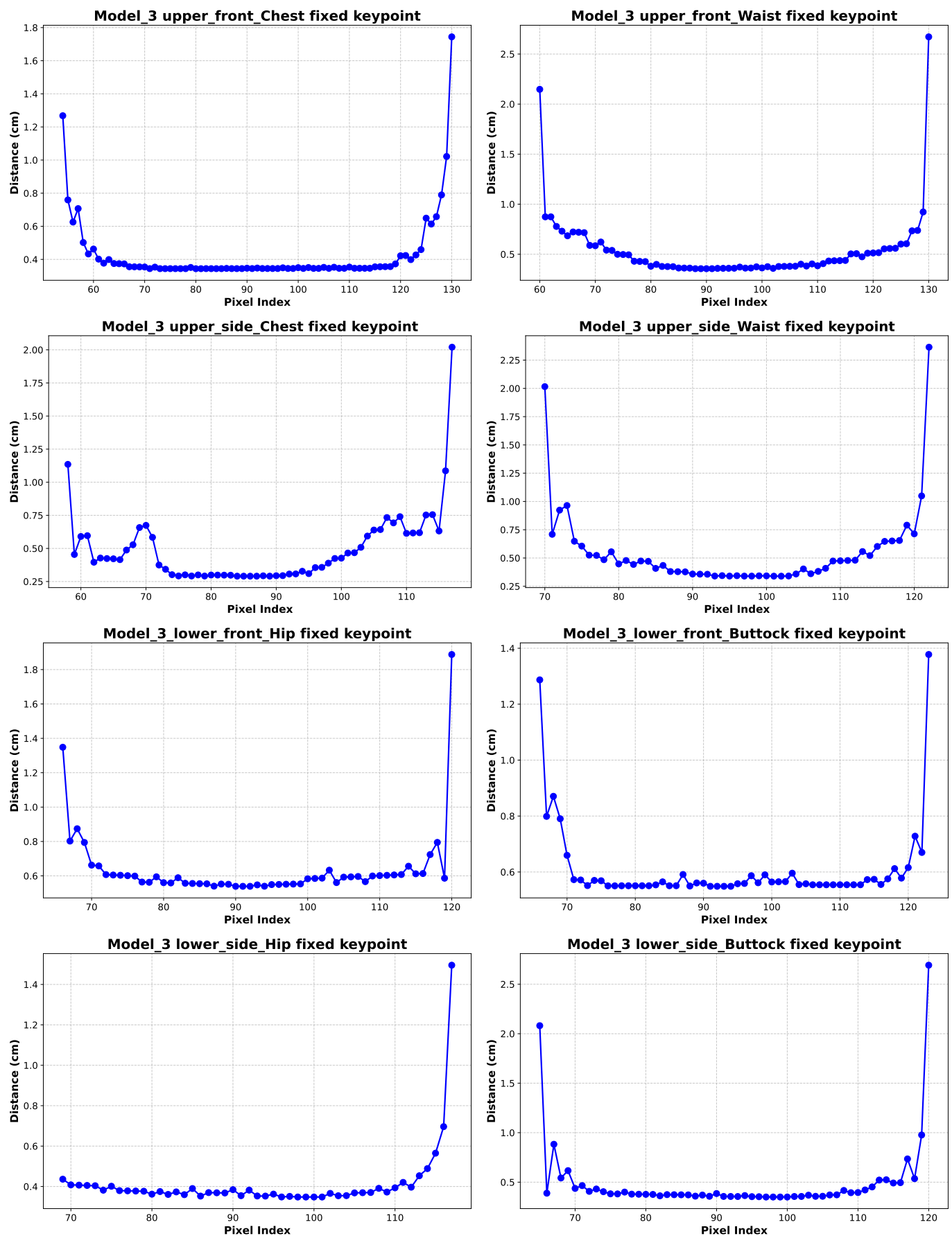


Figure A20. Graphs showing the data after keypoint correction of Model 3. The y-axis automatically adjusts its maximum value as the distance between pixels decreases.

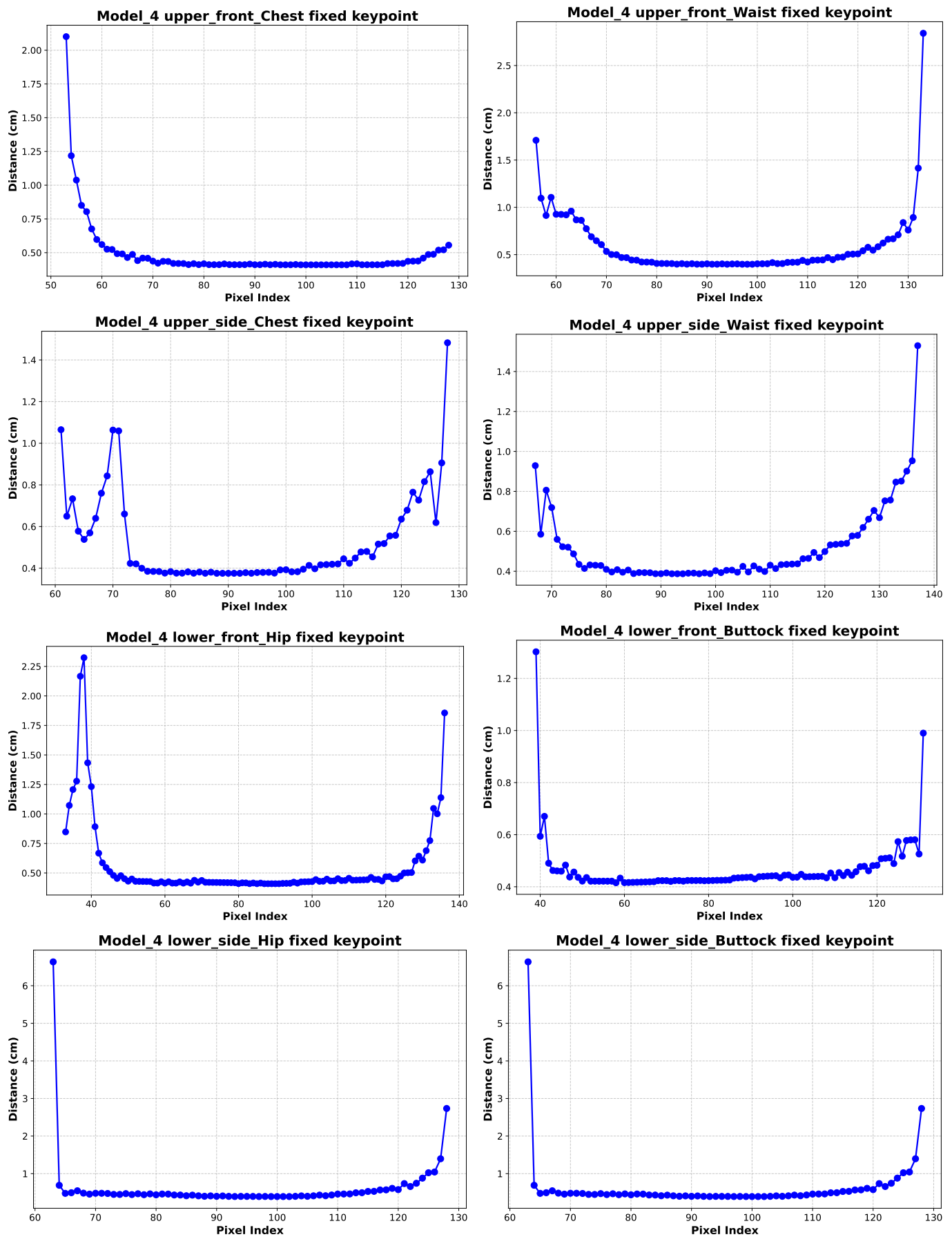


Figure A21. Graphs showing the data after keypoint correction of Model 4. The y-axis automatically adjusts its maximum value as the distance between pixels decreases.

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